

Stakeholder question catalogue

Evidence-led questions for responsible building risk conversations

STAKEHOLDER 36

Insurers and risk assessors

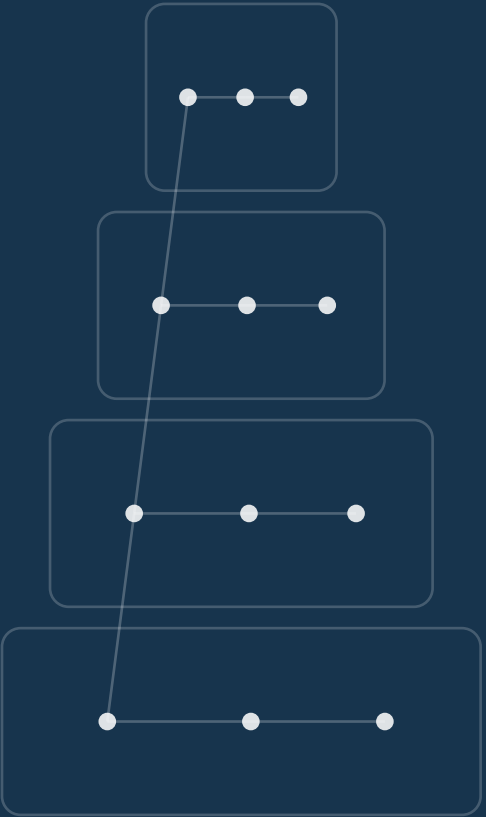
This catalogue helps insurers, risk assessors and accountable building owners frame proportionate evidence requests, scope physical surveys, understand incidents, test resilience assumptions and communicate uncertainty. Every sensor, integration and analytic remains proposed until deployment, mapping, commissioning, permission and current health are independently confirmed.

80 practical questions across eight risk domains

Qualified evidence supports accountable judgement; it does not replace it.

The service does not make underwriting, pricing, coverage or claims decisions.

Policy terms, competent professionals and accountable owners retain authority.



SOURCES

OBSERVATIONS

UNCERTAINTY

PURPOSE

Use the 80 questions to define evidence requests, survey scope, control discussions, incident fact-finding, continuity assessment and accountable escalation. Each card states proposed evidence, authoritative sources, an evidence gate, uncertainty, a stopping boundary and a concrete next action.

AUTHORITY AND PRECEDENCE

Policy terms govern contractual scope. Competent professional findings govern the technical scope they explicitly address. Current accountable-owner records govern declared assets, uses, controls and arrangements. Qualified telemetry supports observed conditions only. Models support bounded reasoning. Conflicts remain visible.

PROPOSED EVIDENCE STATUS

Every sensor, integration and analytic is proposed until deployment, asset and location mapping, commissioning, sensor-specific calibration or functional checks, permissions, clock alignment and current health are independently confirmed. Technical availability alone does not make a signal authoritative.

CONSEQUENTIAL DECISION BOUNDARY

The service must not make or recommend underwriting, pricing, coverage or claims decisions; certify controls, condition or compliance; authorise work, access, occupation or system operation; promise response, resilience, loss prevention or recovery; or replace policy terms, physical surveys and competent judgement.

HISTORICAL SPATIAL REFERENCE

The May 2021 floor plan is historical spatial reference only. It cannot establish current room use, access, services, hazards, compartmentation, occupancy, ownership, restrictions or closures. Every present-day spatial fact needs confirmation from the current owner or a competent professional.

PRIVACY, SECURITY AND FAIRNESS

Use zone-level, aggregated and purpose-limited evidence. Do not identify or track people, infer personal risk, health, intent, negligence or productivity, inspect research, speech or network content, expose restricted records or security layouts, bypass permissions, or combine data for a new purpose without accountable approval.

STAKEHOLDER QUESTION A natural decision-facing question for an insurer or risk assessor to ask an accountable owner. It identifies the asset, zone, hazard or decision period and permits an honest unknown.	SENSORS AND TELEMETRY Physically plausible, privacy-preserving proposed evidence. It states modality and location, spatial and temporal coverage, sampling or aggregation, calibration or functional checks, clock alignment, uptime, gaps, drift and provenance.	AUTHORITATIVE SUPPORTING DATA Named source owners and precedence, effective, observed and retrieved times, permissions, provenance, staleness and conflicts. Authoritative records lead within their stated scope.
ANALYSIS REQUIRED Operation type, decision interval, minimum evidence, method, uncertainty and safe failure. Lookup, observation, calculation, estimate, forecast, comparison, diagnosis and recommendation are distinguished.	ANSWER BOUNDARY What the service must not infer, disclose, promise, certify, authorise, control or decide. Policy terms, competent people and accountable owners retain authority.	WHY THIS IS USEFUL The specific action enabled: request evidence, target a survey, confirm a control, preserve incident material, test a dependency, seek specialist input or limit a statement.

ANSWER STATUS KEY

CONFIRMED Authoritative evidence supports the statement within scope.	INDICATED Qualified telemetry supports an observed condition with limits.	UNKNOWN The minimum evidence or coverage gate is not met.	CONFLICTING Sources disagree and an accountable owner must resolve them.	RESTRICTED Evidence exists but is not disclosable for this purpose.	UNAVAILABLE A source, integration or permission cannot be accessed.
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CATEGORY AND PAGE MAP

1 assets, uses, occupancy context and exposure IR-001-IR-010 pages 4-8	5 inspection, maintenance, compliance and impairments IR-041-IR-050 pages 24-28
2 hazards, vulnerabilities and control ownership IR-011-IR-020 pages 9-13	6 resilience, continuity, dependencies and recovery IR-051-IR-060 pages 29-33
3 fire, security, water, machinery and environmental risks IR-021-IR-030 pages 14-18	7 climate, weather, energy and long-term change IR-061-IR-070 pages 34-38
4 incidents, losses, near misses and claims evidence IR-031-IR-040 pages 19-23	8 evidence quality, uncertainty, scenarios and risk communication IR-071-IR-080 pages 39-43

IR-001

STAKEHOLDER QUESTION

For the current declared exposure, which buildings, zones, fixed plant, portable equipment and contents are evidenced as included, and which items remain unmatched or ambiguous?

SENSORS AND TELEMETRY

Proposed evidence: Asset-tag or barcode inventories, approved BMS point lists, fixed-plant identifiers, submeter-to-equipment associations and zone-level presence checks that do not inspect device or research content. **Coverage/timing:** All declared asset classes and accessible locations; owner-approved snapshots at declaration, after moves or disposals, and at least quarterly, with scan time retained. **Quality gate:** verify ID-to-asset and reader-to-zone maps, scan and aggregation rules, functional checks, clock offset, uptime, failed or duplicate reads, mapping drift and lineage; physical calibration is not applicable. Report counts and owner-supplied value bands by zone; moved, duplicated, untagged, inaccessible or powered-off assets can remain unmatched and a detected tag does not prove ownership or condition.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: The insured-risk schedule custodian, estates asset owner, finance fixed-asset owner, IT or laboratory custodians and competent valuers. Policy schedules lead for contractual declaration, competent valuations for stated value, and current endorsed registers for physical inventory. **Permissions/provenance:** Use purpose-limited joins with source/version identifiers and asset-zone mapping lineage; show every mismatch and route it to the relevant owner rather than silently selecting a record. **Time/staleness/conflicts:** retain register effective, telemetry observed and query retrieved times; treat location or inventory evidence outside the current declaration window as stale; show schedule, register and observation mismatches together.

ANALYSIS REQUIRED

Operation type: lookup + comparison + calculation. **Interval/evidence gate:** At declaration, material change and annual assessment; require one current schedule, owner-approved registers and at least 90% mapped coverage by declared value band. **Method:** Three-way exposure reconciliation and concentration profiling. Keyed joins, duplicate and orphan detection, zone totals and exception sampling against a physical survey. **Uncertainty/safe failure:** State mapped count, mapped value-band share and unmatched share; mark the answer unknown where more than 10% of a material value band is unmapped or valuation dates are unsuitable.

ANSWER BOUNDARY

The service must not infer ownership from proximity, value unrecorded property, set sums insured, decide insurability, pricing, coverage or claim treatment, or treat a tag response as proof of serviceability.

WHY THIS IS USEFUL

Enables the assessor to target a survey, request register correction, clarify the declared schedule and identify exposure concentrations before an accountable insurance decision.

IR-002

STAKEHOLDER QUESTION

Which spaces appear to differ from the current owner-approved use record, and what material change must be confirmed before the declared exposure can be relied on?

SENSORS AND TELEMETRY

Proposed evidence: Anonymous doorway counts, room booking-state indicators, zone power demand and environmental patterns used only as change cues; no audio, video, identities, device content or research activity is inspected. **Coverage/timing:** Mapped rooms and shared areas; 15-minute readings aggregated daily over at least 28 days, plus owner-declared dates for moves, events and temporary occupation. **Quality gate:** validate each counter or presence point against short manual samples; retain entrance coverage, sampling and aggregation, functional checks, clock offset, uptime, occlusion gaps, count drift and raw-to-zone lineage. Similar uses can produce similar signatures, open-plan spillover can distort room evidence and uninstrumented rooms remain unobserved; report only pattern differences, not a new use classification.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: Space management, estates, health and safety, local room custodians, timetabling and event or project owners. Current approved space records and written temporary-use permissions lead; competent fire, safety or planning findings lead within their scope. **Permissions/provenance:** Use only authorised booking and aggregate access metadata with documented transformations; expose stale room records and route declared-versus-observed conflicts to the space owner. **Time/staleness/conflicts:** retain schedule effective, interval observed and query retrieved times; planned states expire at their stated end; observations outside the comparison window are stale, and material differences go to the responsible owner.

ANALYSIS REQUIRED

Operation type: observation + comparison + estimate. **Interval/evidence gate:** Monthly and after a move, project or unusual event; require a current room register, 20 valid comparable days and at least two independent zone-level signals for any indicated change. **Method:** Declared-use reconciliation with change-point screening. Compare owner-declared use periods with robust baselines, change-point tests and rule-based exceptions, then seek owner confirmation. **Uncertainty/safe failure:** Report observed coverage, effect size and a high/medium/low evidence grade; return 'use not established' when evidence is sparse, mixed or seasonally incomparable.

ANSWER BOUNDARY

The service must not identify occupants, infer teaching or research content, determine legal change of use, occupancy class or compliance, authorise occupation, or state that policy terms respond.

WHY THIS IS USEFUL

Enables the assessor to determine whether a competent reassessment, updated declaration, altered survey scope or temporary-control discussion is needed.

IR-003

STAKEHOLDER QUESTION

For each material-risk zone, when is it normally occupied, lightly staffed or apparently unattended, and how reliable is that pattern?

SENSORS AND TELEMETRY

Proposed evidence: Anonymous bidirectional counters at selected entrances, zone presence sensors, door-state counts and lighting or plug-load activity, fused without identities or route reconstruction. **Coverage/timing:** Principal entrances and nominated material-risk zones; five-minute observations aggregated to 30-minute states, with separate weekday, weekend, event and closure profiles. **Quality gate:** validate each counter or presence point against short manual samples; retain entrance coverage, sampling and aggregation, functional checks, clock offset, uptime, occlusion gaps, count drift and raw-to-zone lineage. Tailgating, held-open doors, cleaners, maintenance visits, sensor occlusion and partial entrance coverage can bias counts; absence of signal is never proof that nobody is present.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: Security, facilities duty management, space management, timetabling, event owners and authorised opening-hours custodians. Approved opening, event and closure records lead; security access records lead for recorded access events, while commissioned telemetry supports only observed zone conditions. **Permissions/provenance:** Use least-privilege, aggregated access data and suppress small groups; retain sensor-health and mapping lineage, and present unresolved schedule-versus-observation differences. **Time/staleness/conflicts:** retain schedule effective, interval observed and query retrieved times; planned states expire at their stated end; observations outside the comparison window are stale, and material differences go to the responsible owner.

ANALYSIS REQUIRED

Operation type: observation + estimate. **Interval/evidence gate:** Weekly profiling and event-specific assessment; require four representative weeks, at least 85% valid intervals and coverage of every entrance used in the estimate. **Method:** Multi-signal occupancy-state estimation and unattended-window detection. Calendar reconciliation followed by probabilistic state fusion, with cross-validation against manually confirmed sample periods. **Uncertainty/safe failure:** Publish false-empty and false-occupied estimates from validation samples and confidence by time band; default to 'presence unknown' whenever coverage or validation gates fail.

ANSWER BOUNDARY

The service must not identify or track people, infer lone-worker identity or behaviour, promise security response, authorise opening, or use an estimated unattended state as a factual attendance record.

WHY THIS IS USEFUL

Supports survey timing, after-hours control checks, response-planning discussions and evidence requests for zones that are frequently unattended.

IR-004

STAKEHOLDER QUESTION

Where are high-value, irreplaceable or continuity-critical assets concentrated, and which single peril zone or shared dependency could affect several together?

SENSORS AND TELEMETRY

Proposed evidence: Owner-approved asset-zone mapping, local leak detection, temperature and humidity sensing, power-state telemetry and verified fire or drainage zone references; no content from protected equipment is collected. **Coverage/timing:** All owner-declared critical assets and the rooms, cabinets or plant areas containing them; condition data at one- to fifteen-minute intervals and location snapshots after every recorded move. **Quality gate:** verify ID-to-asset and reader-to-zone maps, scan and aggregation rules, functional checks, clock offset, uptime, failed or duplicate reads, mapping drift and lineage; physical calibration is not applicable. Location accuracy is limited by tag mapping, shared rooms and mobile assets; condition sensors observe only their commissioned area and cannot establish replacement value or criticality.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: Asset custodians, finance, estates, IT service owners, research or teaching service owners and business-continuity leads. Competent valuations and owner-approved criticality records lead, followed by current asset registers and verified location records; telemetry supports only the observed condition and period. **Permissions/provenance:** Restrict detailed locations for security-sensitive assets, retain source/version and zone lineage, and show conflicts between value, criticality and location records to their owners. **Time/staleness/conflicts:** retain register effective, telemetry observed and query retrieved times; treat location or inventory evidence outside the current declaration window as stale; show schedule, register and observation mismatches together.

ANALYSIS REQUIRED

Operation type: lookup + calculation + comparison. **Interval/evidence gate:** At least annually and after material moves; require current criticality and value bands, at least 90% mapped declared value, and verified hazard-zone boundaries for the peril being considered. **Method:** Spatial accumulation and common-cause exposure analysis. Aggregate by room, compartment, drainage catchment, power source and environmental-control dependency, then test single-zone and single-service loss scenarios. **Uncertainty/safe failure:** Report exposed value and criticality as ranges with unmapped shares; do not rank zones when more than 15% of the relevant value or dependency mapping is uncertain.

ANSWER BOUNDARY

The service must not calculate or present a probable maximum loss, invent values, reveal restricted asset locations, decide coverage or recommend a premium, or treat co-location as proof of common damage.

WHY THIS IS USEFUL

Enables a focused physical survey, separation or protection discussion, asset-register correction and continuity assessment for concentrated exposures.

IR-005

STAKEHOLDER QUESTION

Which recurring events or operating peaks temporarily increase occupancy, electrical demand, combustibles, deliveries or access pressure beyond the normal profile?

SENSORS AND TELEMETRY

Proposed evidence: Approved event and booking indicators, anonymous entrance counts, aggregate electrical demand, temporary distribution-board monitoring, door-cycle counts and owner-declared temporary-storage checks. **Coverage/timing:** Event spaces, access routes and affected service zones; five-minute signals aggregated by event phase, with setup, live, close-down and overnight periods retained separately. **Quality gate:** validate each counter or presence point against short manual samples; retain entrance coverage, sampling and aggregation, functional checks, clock offset, uptime, occlusion gaps, count drift and raw-to-zone lineage. Electrical demand does not identify equipment, counts may miss entrances and temporary combustibles usually require inspection; compare only events with similar duration, season and operating mode.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: Event owners, timetabling, estates operations, security, fire-safety and temporary-power or contractor coordinators. Approved event plans, occupancy limits, permits and competent findings lead; observed telemetry can test stated timing and aggregate loading only. **Permissions/provenance:** Use event identifiers rather than attendee identities, document source joins and exclude restricted details; show any plan-versus-observation exception to the event and control owners. **Time/staleness/conflicts:** retain schedule effective, interval observed and query retrieved times; planned states expire at their stated end; observations outside the comparison window are stale, and material differences go to the responsible owner.

ANALYSIS REQUIRED

Operation type: observation + comparison. **Interval/evidence gate:** Before high-consequence events and quarterly for recurring events; require an approved plan, mapped affected zones, healthy signals for at least 90% of event duration and three comparable events where trend claims are made. **Method:** Event-phase exposure profiling and peak exception detection. Normalise counts, demand and door cycles by event duration and area, compare percentiles with the approved plan and identify concurrent temporary controls or impairments. **Uncertainty/safe failure:** Show observed peak ranges and data loss; use a single-event statement only as a bounded observation and return 'not comparable' when event types or coverage differ materially.

ANSWER BOUNDARY

The service must not identify attendees, infer event content or behaviour, certify occupancy or fire load, approve an event, or decide whether the exposure is acceptable under policy terms.

WHY THIS IS USEFUL

Helps scope pre-event checks, temporary protection, security staffing discussions, specialist attendance and post-event evidence requests.

IR-006

STAKEHOLDER QUESTION

Which declared hazardous, temperature-sensitive, irreplaceable or theft-attractive materials are present, where are they stored, and is their required storage envelope evidenced?

SENSORS AND TELEMETRY

Proposed evidence: Enclosure or room temperature and humidity, leak and water-on-floor sensors, door-state counts, local power availability and approved storage-alarm status; no chemical, specimen or research content is sensed. **Coverage/timing:** Only owner-nominated storage rooms, cabinets and supporting services; one- to five-minute readings with hourly summaries and immediate recording of threshold excursions or power loss. **Quality gate:** confirm representative siting, asset or zone mapping, units, traceable calibration or functional check, sampling and aggregation, clock offset, uptime, gaps, response lag, drift and raw-to-summary lineage. A room sensor may not represent cabinet conditions, door opening can cause short excursions and sensor drift can mimic change; unmonitored stores and unverified set-points remain unknown.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: Health and safety, hazardous-materials or biological-safety owners, local custodians, security, estates and the asset or collections owner. Current approved inventories, storage specifications, safety documentation and competent findings lead; telemetry supports the observed environment only. **Permissions/provenance:** Minimise or withhold sensitive material identities and exact locations, use coded storage references, retain calibration and mapping provenance, and route conflicts to the competent owner. **Time/staleness/conflicts:** retain register effective, telemetry observed and query retrieved times; treat location or inventory evidence outside the current declaration window as stale; show schedule, register and observation mismatches together.

ANALYSIS REQUIRED

Operation type: lookup + observation + comparison. **Interval/evidence gate:** Daily for critical stores and after alarms or moves; require a current owner-approved requirement, verified sensor location, seven valid days for baseline claims and at least 95% interval completeness. **Method:** Storage-declaration reconciliation and environmental-envelope assessment. Compare measurements with the specified envelope using excursion duration, degree-time and service-state context rather than isolated threshold crossings. **Uncertainty/safe failure:** Report sensor uncertainty, coverage and excursion ranges; mark the condition unknown when cabinet representativeness, calibration or the authoritative requirement is unverified.

ANSWER BOUNDARY

The service must not disclose restricted inventories or exact locations, infer material identity, certify storage safety, determine regulatory compliance, control the store, or decide insurance response.

WHY THIS IS USEFUL

Enables the assessor to request a competent inspection, clarify declared stock, assess protection dependencies and preserve evidence of environmental excursions.

IR-007

STAKEHOLDER QUESTION

What temporary exposure is created by active construction, refurbishment, commissioning, relocation or intrusive maintenance, and during which zones and time windows?

SENSORS AND TELEMETRY

Proposed evidence: Approved permit-state feeds, temporary temperature, dust, water, differential-pressure or power monitoring, fire-system impairment indicators and door or barrier state at the defined work area. **Coverage/timing:** The mapped work zone, adjacent affected compartments and services; one- to five-minute observations during work and aftercare, plus daily owner-confirmed start, stop and hand-back times. **Quality gate:** for every contributing stream, verify asset and location mapping, units, calibration or functional check, sampling and aggregation, clock offset, uptime, gaps, drift and source-to-result lineage; exclude any stream that fails its gate. Temporary sensors are easily moved or damaged, permit status does not prove field compliance and some hazards require physical observation; unmapped work or undocumented changes cannot be analysed reliably.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: Project sponsor, principal contractor, permit issuer, estates, fire-safety, health and safety, security and affected local custodians. Signed permits, approved method statements, isolation schedules and competent site findings lead; telemetry can corroborate stated conditions only within its mapped area and time. **Permissions/provenance:** Access contractor and security details only for the stated purpose, retain permit/version and sensor-placement provenance, and expose disagreement between planned, observed and hand-back states. **Time/staleness/conflicts:** retain schedule effective, interval observed and query retrieved times; planned states expire at their stated end; observations outside the comparison window are stale, and material differences go to the responsible owner.

ANALYSIS REQUIRED

Operation type: lookup + observation + comparison. **Interval/evidence gate:** Before work, each shift and at hand-back; require a current permit or written scope, mapped affected systems, named control owners and healthy evidence covering at least 90% of the higher-risk window. **Method:** Time-bounded works exposure and control-overlap assessment. Construct a timeline of work phases, isolations, temporary controls, alarms and environmental excursions, then identify periods with missing or overlapping safeguards. **Uncertainty/safe failure:** Quantify uncovered minutes, unmapped services and evidence gaps; stop and escalate when a critical isolation, permit, temporary control or competent owner cannot be confirmed.

ANSWER BOUNDARY

The service must not approve permits, authorise work or reoccupation, direct contractors, certify temporary controls, infer negligence, or state that an insurer will accept the temporary exposure.

WHY THIS IS USEFUL

Supports targeted site attendance, evidence requests, temporary-protection discussion and confirmation that hand-back evidence is complete.

IR-008

STAKEHOLDER QUESTION

Which neighbouring premises, campus services or external activities could materially increase exposure, and which controls sit outside the building owner's authority?

SENSORS AND TELEMETRY

Proposed evidence: Shared-utility status, boundary water-level or drainage indicators, external weather, common access or gate status and building-interface alarms; no surveillance of neighbouring people or activities. **Coverage/timing:** Verified shared feeds, boundary interfaces and nominated external exposure points; one- to fifteen-minute operational data with event snapshots and owner-confirmed interface maps reconfirmed after change. **Quality gate:** verify station or model location, exposure and representativeness; retain units, quality flags, calibration or bias checks, observation or issue time, sampling and aggregation, clock basis, uptime, missing periods, drift and source-model lineage. External assets may be uninstrumented, data licences may limit detail and correlation does not establish third-party causation; the service cannot observe beyond agreed boundaries.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: Campus estates, utility and network service owners, landlords or tenants, neighbouring responsible parties, security and emergency-planning owners. Leases, service agreements, easements, competent surveys and current interface records lead; external data retains the source owner's effective and observation times plus the service retrieval time. **Permissions/provenance:** Honour third-party licences and confidentiality, store only necessary interface data, retain source and transformation lineage, and present unresolved responsibility or boundary conflicts without attribution. **Time/staleness/conflicts:** retain contract or source effective, external observed and local retrieved times; expired or unverifiable third-party assertions are unknown; preserve differences with local records and identify the external and internal owners.

ANALYSIS REQUIRED

Operation type: lookup + comparison + estimate. **Interval/evidence gate:** At annual assessment, after interface change and during relevant alerts; require current responsibility maps, an owner for each material interface and verified coverage of at least 90% of critical shared feeds. **Method:** External-interface exposure mapping and dependency exception analysis. Link interfaces to affected zones and controls, compare current state with contractual service assumptions and test bounded failure or ingress scenarios. **Uncertainty/safe failure:** State observed coverage and dependency uncertainty; return 'external condition not evidenced' when the external source, licence, boundary or current owner cannot be verified.

ANSWER BOUNDARY

The service must not infer third-party intent, fault or negligence, disclose restricted security arrangements, promise utility performance, interpret legal responsibility, or decide recovery under policy terms.

WHY THIS IS USEFUL

Helps identify whom to contact, which agreements or surveys to request, and where a joint inspection or contingency discussion is justified.

IR-009

STAKEHOLDER QUESTION

Which rarely visited roofs, basements, risers, stores and plant areas have the longest blind intervals for developing loss conditions?

SENSORS AND TELEMETRY

Proposed evidence: Proposed leak, sump-level, temperature, humidity, vibration, door-open and local power-health sensors, together with inspection check-in timestamps that record attendance but not identity. **Coverage/timing:** All owner-designated concealed or low-visit spaces that can be safely instrumented; one- to five-minute event sensing, hourly summaries and monthly verification of placement, battery, communications and access. **Quality gate:** confirm sensor mounting, asset identity, units and load context; retain calibration or functional checks, sample rate and aggregation, clock offset, uptime, gaps, mounting change, drift and feature lineage. Wireless shadowing, battery depletion, blocked sensing surfaces and inaccessible areas create blind spots; a healthy sensor covers only its commissioned point and cannot replace a competent inspection.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: Estates maintenance, roof and drainage owners, fire-safety, security, local custodians and the telemetry service owner. Current asset and inspection records lead for area ownership and required checks; verified telemetry supports observed conditions only. **Permissions/provenance:** Use coded locations where security requires, retain commissioning and health provenance, and show unmapped or inaccessible areas as coverage gaps rather than treating them as normal. **Time/staleness/conflicts:** retain schedule effective, interval observed and query retrieved times; planned states expire at their stated end; observations outside the comparison window are stale, and material differences go to the responsible owner.

ANALYSIS REQUIRED

Operation type: observation + comparison. **Interval/evidence gate:** Monthly and before seasonal risk assessments; require a current list of material low-visit areas, mapped inspection frequency and verified health for at least 90% of proposed monitoring points. **Method:** Observation-coverage gap analysis and hidden-condition screening. Score each area by visit interval, sensor coverage, hazard consequence and recent anomalies, then validate the highest gaps through owner-led inspection. **Uncertainty/safe failure:** Publish the proportion of risk areas observed, median evidence age and longest blind interval; do not rank an area as low risk solely because no anomaly was recorded.

ANSWER BOUNDARY

The service must not authorise access to restricted spaces, infer safe condition from silence, replace roof, confined-space or electrical inspections, disclose sensitive layouts, or control alarms or plant.

WHY THIS IS USEFUL

Enables proportionate inspection targeting and monitoring proposals for places where leaks, overheating, security breaches or plant deterioration could otherwise remain hidden.

IR-010

STAKEHOLDER QUESTION

Where do declared assets or activities accumulate behind a common fire, water, power, cooling, security, network or access dependency?

SENSORS AND TELEMETRY

Proposed evidence: Owner-approved asset locations, verified distribution-board and submeter associations, water and cooling branch status, fire-zone references, access-control availability and network or gateway service health. **Coverage/timing:** All declared critical assets and nominated common services; state changes at one- to fifteen-minute intervals, with topology snapshots whenever circuits, valves, zones or gateways are altered. **Quality gate:** verify the current asset-zone-service topology with owner sign-off; retain version and effective time, mapping completeness, geometry precision, timestamp alignment, source availability, change drift and lineage. Calibration is not applicable to topology. Building service diagrams can be stale, telemetry may show state without full topology and shared dependencies can be hidden behind local controllers; mobile assets and unmapped branches remain outside totals.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: Estates engineering, fire-safety, utilities, IT, security, asset custodians, finance and business-continuity owners. Competent schematics, protection studies and current owner-approved topology lead; asset registers lead for declared exposure; telemetry supports current state only. **Permissions/provenance:** Restrict detailed security and network topology, retain node-edge-source lineage, and expose conflicts between drawings, registers and observed service state to the responsible engineering owner. **Time/staleness/conflicts:** retain topology effective or version, supporting observed and query retrieved times; a topology becomes stale after material change; show drawing, register and telemetry differences without selecting a winner.

ANALYSIS REQUIRED

Operation type: lookup + calculation + comparison. **Interval/evidence gate:** At annual assessment and after material service changes; require validated topology for every material dependency, at least 90% of declared critical assets mapped and a current owner for each graph edge. **Method:** Dependency-graph accumulation and common-cause scenario analysis. Build a typed dependency graph, calculate exposure reachable from single-node and selected multi-node failures, and identify cut sets requiring competent validation. **Uncertainty/safe failure:** Report exposure ranges, unmapped value or criticality shares and topology confidence; return 'accumulation not quantifiable' where critical edges or assets are materially unverified.

ANSWER BOUNDARY

The service must not present a maximum-loss estimate, assume simultaneous damage from shared service, reveal restricted topology, decide policy accumulation, or recommend underwriting, pricing or coverage action.

WHY THIS IS USEFUL

Supports survey scoping, separation and redundancy discussions, correction of dependency records and accountable assessment of concentrated common-cause exposure.

IR-011

STAKEHOLDER QUESTION

For each material zone, which hazards and residual vulnerabilities are current, and who owns acceptance, treatment, evidence and escalation?

SENSORS AND TELEMETRY

Proposed evidence: Hazard-specific supporting signals such as leak, temperature, door, fire-interface, plant-condition, utility and environmental status, mapped only to the zone and control they can physically observe. **Coverage/timing:** Owner-designated material hazards and affected zones; event-level observations with daily summaries, plus snapshots at the effective date of every formal risk-register change. **Quality gate:** for every contributing stream, verify asset and location mapping, units, calibration or functional check, sampling and aggregation, clock offset, uptime, gaps, drift and source-to-result lineage; exclude any stream that fails its gate. Telemetry can indicate a condition or control state but cannot establish hazard likelihood, consequence or residual risk; uninstrumented hazards and latent construction defects remain outside observation.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: The institutional risk owner, health and safety, estates, fire, security, business-continuity, local activity owners and competent specialist assessors. Current competent risk assessments and signed risk registers lead within scope, followed by accountable-owner records; telemetry supports only the stated observed condition and time. **Permissions/provenance:** Limit access by hazard and role, retain assessment version, acceptance date, zone mapping and transformation lineage, and show owner or assessment conflicts for accountable resolution. **Time/staleness/conflicts:** retain owner-record effective, event observed and query retrieved times; live status is stale after a missed heartbeat or 15 minutes unless the owner sets a stricter limit; preserve zone and time disagreements.

ANALYSIS REQUIRED

Operation type: lookup + comparison. **Interval/evidence gate:** At each formal risk assessment and material change; require a current hazard register, named accountable owner, stated control set and current evidence for at least 90% of controls linked to material hazards. **Method:** Hazard-control-owner reconciliation with residual-evidence grading. Build a hazard-by-zone matrix, link preventive, detective, containment and response controls, and compare declared residual exposure with verified control evidence. **Uncertainty/safe failure:** Report control-evidence coverage, age and a high/medium/low residual-evidence grade rather than a risk score; return unknown where hazard definition, consequence basis or ownership is missing.

ANSWER BOUNDARY

The service must not create or accept residual risk, infer personal vulnerability, certify controls, determine legal duty, replace a competent assessment, or make underwriting, pricing or coverage decisions.

WHY THIS IS USEFUL

Gives the assessor a defensible map of which hazards need owner clarification, specialist survey, updated evidence or an accountable treatment discussion.

IR-012

STAKEHOLDER QUESTION

For each material hazard, is the prevention-detection-containment-response chain complete, and which link is missing, impaired or unverified?

SENSORS AND TELEMETRY

Proposed evidence: Verified status from relevant alarms, detectors, valves, pumps, fire doors, power supplies, environmental controls and notification gateways, using only state and health data needed for the named barrier. **Coverage/timing:** Every owner-declared barrier in selected material hazard chains; seconds-to-minutes for state changes, daily health summaries and records of scheduled functional tests. **Quality gate:** verify point-to-asset or zone mapping and event-code semantics; test contacts, alarms and heartbeats; retain event aggregation, clock offset, uptime, lost events, configuration drift and source-to-summary lineage. Calibrate only underlying physical sensors. A normal status bit does not prove field performance, interfaces may fail silently and manual barriers may have no telemetry; physical condition and competent testing remain necessary.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: The hazard owner, estates engineering, fire-safety, security, health and safety, IT or communications owners and emergency-response coordinators. Competent designs, cause-and-effect documents, test findings and current owner-approved control registers lead; commissioned telemetry supports observed availability only. **Permissions/provenance:** Preserve barrier IDs, upstream source, mapping and test lineage; restrict sensitive security or life-safety detail and leave contradictory states visible for the competent control owner. **Time/staleness/conflicts:** retain owner-record effective, event observed and query retrieved times; live status is stale after a missed heartbeat or 15 minutes unless the owner sets a stricter limit; preserve zone and time disagreements.

ANALYSIS REQUIRED

Operation type: lookup + comparison + diagnosis. **Interval/evidence gate:** Quarterly, after impairment and before a material survey; require an owner-approved hazard-control map, current test evidence and verified status for at least 95% of critical automated barriers. **Method:** Bow-tie barrier-chain completeness and availability assessment. Classify barriers by function, independence and common dependency, then identify missing links, stale tests, simultaneous impairments and unsupported manual assumptions. **Uncertainty/safe failure:** Publish evidenced-barrier share, unknown barriers and common-dependency count; safe-fail to 'chain not established' when a critical barrier, test or owner is absent.

ANSWER BOUNDARY

The service must not certify barrier adequacy, infer that a hazard is controlled, authorise operation with a missing barrier, expose restricted cause-and-effect details, or decide insurance acceptance.

WHY THIS IS USEFUL

Supports a focused control survey, evidence request and discussion with the owner about missing, dependent or poorly evidenced layers of protection.

STAKEHOLDER QUESTION

IR-013

Which control, utility or service component is a credible single point of failure across multiple hazards, zones or high-consequence assets?

SENSORS AND TELEMETRY

Proposed evidence: Service-state telemetry from distribution boards, valves, pumps, gateways, controllers, UPS units, cooling branches, communication links and shared alarm interfaces, tied to verified topology. **Coverage/timing:** Owner-nominated critical services and their downstream zones; state changes at one- to fifteen-minute intervals and topology snapshots after configuration or project changes. **Quality gate:** verify the current asset-zone-service topology with owner sign-off; retain version and effective time, mapping completeness, geometry precision, timestamp alignment, source availability, change drift and lineage. Calibration is not applicable to topology. Status data may omit hidden physical dependencies, redundant components may share upstream supplies and drawings can lag reality; an apparent duplicate path is not proof of independence.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: Estates engineering, utilities, fire and security system owners, IT, asset custodians and business-continuity owners. Competent schematics, protection studies and owner-approved dependency records lead; telemetry shows only observed component state. **Permissions/provenance:** Restrict detailed topology, preserve node-edge-source lineage and route drawing-versus-observation conflicts to the engineering owner rather than auto-correcting the graph. **Time/staleness/conflicts:** retain topology effective or version, supporting observed and query retrieved times; a topology becomes stale after material change; show drawing, register and telemetry differences without selecting a winner.

ANALYSIS REQUIRED

Operation type: lookup + comparison + estimate. **Interval/evidence gate:** At annual risk assessment and after material change; require current topology for all critical services, verified ownership for every material node and at least 90% mapped downstream criticality. **Method:** Fault-tree, cut-set and common-dependency analysis. Construct fault trees and dependency graphs, identify minimal cut sets and test selected component-failure scenarios against recorded failover or exercise evidence. **Uncertainty/safe failure:** Show topology confidence, unmapped downstream share and scenario range; do not label a component a single point when independence or alternative paths remain unverified.

ANSWER BOUNDARY

The service must not disclose sensitive network or security architecture, guarantee redundancy, control failover, certify design resilience, or translate a cut set into an underwriting or pricing decision.

WHY THIS IS USEFUL

Helps prioritise engineering verification, redundancy surveys, contingency planning and evidence requests around potentially disproportionate failure points.

STAKEHOLDER QUESTION

IR-014

What has changed since the latest competent assessment, and which findings, drawings or assumptions now require revalidation?

SENSORS AND TELEMETRY

Proposed evidence: Approved configuration-change records, BMS point additions or removals, asset-move indicators, space-use cues, project isolation states and aggregate occupancy or load shifts. **Coverage/timing:** The assessment's stated scope and all zones or systems changed since its effective date; daily change summaries and event timestamps linked to project, move or configuration records. **Quality gate:** where telemetry is not applicable, validate identifiers, scope, status codes, signatures, timestamp and time-zone alignment, retrieval completeness, missing records, source availability, schema drift and transformation lineage; mark calibration and sampling not applicable. Automated logs miss unrecorded physical work, a changed signal does not reveal technical significance and historical floor-plan geometry cannot establish current use or systems.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: The competent assessor, estates change-control owner, space management, project managers, fire and safety owners, security and local custodians. The signed assessment defines its original scope; current approved change records and competent post-change findings lead for later conditions. **Permissions/provenance:** Use permissioned change metadata with version lineage, distinguish planned from observed changes and show any unrecorded or conflicting change to the accountable owner. **Time/staleness/conflicts:** retain schedule effective, interval observed and query retrieved times; planned states expire at their stated end; observations outside the comparison window are stale, and material differences go to the responsible owner.

ANALYSIS REQUIRED

Operation type: lookup + comparison. **Interval/evidence gate:** Before relying on an older assessment and after each material project; require the original assessment, a complete change window and owner confirmation for at least 90% of detected material configuration changes. **Method:** Assessment-baseline delta and assumption-impact screening. Compare documented baseline, current registers and observed configurations, classify each delta by affected hazard, control, exposure and original assumption, then refer material items for competent assessment. **Uncertainty/safe failure:** Report confirmed, indicated and unexplained changes with coverage percentages; do not extend an old conclusion across a materially changed or unobserved scope.

ANSWER BOUNDARY

The service must not declare a competent assessment invalid or current, infer technical significance without its owner, establish present room use from the May 2021 plan, or certify the changed condition.

WHY THIS IS USEFUL

Enables the assessor to decide which earlier conclusions can be used cautiously and where a refreshed survey or specialist confirmation is necessary.

STAKEHOLDER QUESTION

IR-015

Where do control responsibilities transfer among the building owner, occupiers, university teams, contractors and service providers, and are those hand-offs explicit?

SENSORS AND TELEMETRY

Proposed evidence: Workflow evidence such as alarm-routing tests, work-order assignment, acknowledgement, permit acceptance, duty-rota coverage and service hand-off timestamps; physical sensing is used only for the linked control state. **Coverage/timing:** All material controls with more than one organisation or team involved; event-level routing records and monthly snapshots of named accountable and operational roles. **Quality gate:** where telemetry is not applicable, validate identifiers, scope, status codes, signatures, timestamp and time-zone alignment, retrieval completeness, missing records, source availability, schema drift and transformation lineage; mark calibration and sampling not applicable. An assignment or acknowledgement does not prove competence, attendance or completed action; informal practices and emergency substitutions may not appear in systems.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: Building and institutional risk owners, occupiers, estates, security, fire and safety teams, IT, procurement, contractors and service providers. Leases, contracts, service schedules, statutory appointments and signed responsibility matrices lead; operational systems support observed assignment and acknowledgement only. **Permissions/provenance:** Minimise personal data by using role identifiers, preserve contract and workflow provenance, and show gaps, overlaps or conflicting ownership without assigning legal liability. **Time/staleness/conflicts:** retain contract or source effective, external observed and local retrieved times; expired or unverifiable third-party assertions are unknown; preserve differences with local records and identify the external and internal owners.

ANALYSIS REQUIRED

Operation type: lookup + comparison. **Interval/evidence gate:** At contract start, annual assessment and organisational change; require a current responsibility source and named accountable, responsible and escalation roles for every material control. **Method:** Responsibility-interface and escalation-path analysis. Construct a control-by-role RACI and event-routing matrix, identify unowned steps, duplicate authority, expired contacts and hand-offs that exceed owner-approved time limits. **Uncertainty/safe failure:** Report the share of controls with complete ownership and tested routing; return 'owner unresolved' where documents conflict or a role cannot be confirmed.

ANSWER BOUNDARY

The service must not determine legal liability, infer individual performance or negligence, disclose personal contact details beyond permission, appoint duty holders, or authorise actions on another owner's behalf.

WHY THIS IS USEFUL

Enables the assessor to direct evidence requests and recommendations to the correct accountable party and expose dangerous ownership gaps before an incident.

STAKEHOLDER QUESTION

IR-016

Which open defects or risk treatments have the strongest credible bearing on material loss exposure, and how long have they remained unresolved?

SENSORS AND TELEMETRY

Proposed evidence: Work-order state, linked control status, repeat fault codes, leak or environmental excursions, isolation indicators and completion evidence references, without monitoring worker behaviour. **Coverage/timing:** All open actions linked to material hazards, controls or critical assets; daily status snapshots, event timestamps and age measured from the authoritative issue date, not from data ingestion. **Quality gate:** where telemetry is not applicable, validate identifiers, scope, status codes, signatures, timestamp and time-zone alignment, retrieval completeness, missing records, source availability, schema drift and transformation lineage; mark calibration and sampling not applicable. Ticket priority may be inconsistent, closure codes can be premature and telemetry cannot establish the quality of remedial work; actions without verified asset or hazard mapping remain unranked.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: The competent finding owner, estates maintenance, fire, security, health and safety, project owners, procurement and local asset custodians. Competent findings and signed action requirements lead, followed by accountable-owner defect and work-order records; telemetry supports current observed condition only. **Permissions/provenance:** Preserve action, asset, hazard and evidence lineage, restrict commercially sensitive contractor detail and show conflicting dates, priorities or closure states to the action owner. **Time/staleness/conflicts:** retain action effective or version, linked observed and query retrieved times; status without a dated owner update or acceptance evidence is stale; keep open and closed records unresolved until the accountable owner confirms the state.

ANALYSIS REQUIRED

Operation type: lookup + comparison. **Interval/evidence gate:** Weekly for high-consequence actions and monthly otherwise; require a named owner, hazard or control link, due date, current state and evidence reference for at least 95% of material open actions. **Method:** Risk-linked backlog ageing and evidence-based triage. Group by hazard and control, calculate age and overdue duration, identify repeated deferral and overlay current condition indicators without converting them into an automatic risk score. **Uncertainty/safe failure:** Show count, age bands, evidence completeness and sensitivity to owner-assigned consequence class; safe-fail to unranked when consequence, due date or asset mapping is absent.

ANSWER BOUNDARY

The service must not instruct maintenance, override owner priorities, infer contractor fault, certify closure, quantify insured loss, or make pricing, coverage or claims decisions from a backlog ranking.

WHY THIS IS USEFUL

Supports a proportionate evidence request, targeted survey and accountable discussion about overdue treatments that may materially change the risk view.

IR-017

STAKEHOLDER QUESTION

Which permanent controls are impaired or unavailable now, what temporary measures are formally active, and when must those measures be checked or withdrawn?

SENSORS AND TELEMETRY

Proposed evidence: Verified isolation, bypass, fault and out-of-service states from relevant controls, plus temporary detector, patrol, pump, barrier or notification health where the owner has approved monitoring. **Coverage/timing:** The full affected control, protected zones and temporary-measure period; event-level state changes, daily reconciliations and explicit start, recheck and expiry timestamps. **Quality gate:** verify point-to-asset or zone mapping and event-code semantics; test contacts, alarms and heartbeats; retain event aggregation, clock offset, uptime, lost events, configuration drift and source-to-summary lineage. Calibrate only underlying physical sensors. A normal panel status does not prove field function, temporary manual measures may be only partly observable and undocumented impairments remain invisible.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: The control owner, estates duty manager, fire-safety, security, health and safety, project or contractor owner and business-continuity lead. Signed impairment declarations, permits and competent findings lead; owner-approved temporary-control records follow; commissioned telemetry supports observed state only. **Permissions/provenance:** Restrict sensitive control layouts, retain impairment and temporary-measure lineage, and show inconsistent scope, timing or status for immediate owner resolution. **Time/staleness/conflicts:** retain owner-record effective, event observed and query retrieved times; live status is stale after a missed heartbeat or 15 minutes unless the owner sets a stricter limit; preserve zone and time disagreements.

ANALYSIS REQUIRED

Operation type: lookup + observation + comparison. **Interval/evidence gate:** Continuous for declared critical impairments and at each shift handover; require a named owner, affected scope, start time, recheck or expiry, restoration criterion and evidence for every stated temporary measure. **Method:** Impairment-window reconciliation and compensating-control coverage analysis. Align permanent-control state with impairment records and temporary-measure evidence, calculate uncovered minutes and identify expired or unacknowledged arrangements. **Uncertainty/safe failure:** Report observed coverage and clock alignment; stop and escalate when a critical impairment lacks an owner, end point, required temporary control or reliable current state.

ANSWER BOUNDARY

The service must not decide that temporary measures are adequate, authorise continued operation, certify restoration, control systems, expose restricted protection details, or state that insurance acceptance continues.

WHY THIS IS USEFUL

Enables prompt owner contact, specialist attendance, survey reprioritisation and clear documentation of the exact impaired period.

IR-018

STAKEHOLDER QUESTION

Are verified precursor conditions - such as minor leaks, overheating, forced doors, abnormal cycling or unexplained shutdowns - increasing in rate, duration or severity?

SENSORS AND TELEMETRY

Proposed evidence: Mapped leak, temperature, door-forced, pump-cycle, vibration, electrical-current and fault-state signals, using event metadata rather than identities or system content. **Coverage/timing:** Owner-nominated vulnerable assets and zones; event-level capture with daily counts, 30- or 90-day rolling summaries and separate operating-state or seasonal context. **Quality gate:** for every contributing stream, verify asset and location mapping, units, calibration or functional check, sampling and aggregation, clock offset, uptime, gaps, drift and source-to-result lineage; exclude any stream that fails its gate. Sensor thresholds can change, repeated alarms can arise from one unresolved fault and maintenance or occupancy changes can alter baselines; cross-asset comparisons require comparable sensors and settings.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: Estates maintenance, security, fire and safety control owners, incident management, local custodians and the telemetry owner. Incident and maintenance records lead for confirmed events and causes; owner-approved thresholds define recorded precursors; telemetry provides observations only, preserving effective, observed and retrieval times. **Permissions/provenance:** Retain threshold version, asset mapping and deduplication rules, suppress personal access detail and show conflicts between alarms, tickets and owner classifications. **Time/staleness/conflicts:** retain owner-record effective, event observed and query retrieved times; live status is stale after a missed heartbeat or 15 minutes unless the owner sets a stricter limit; preserve zone and time disagreements.

ANALYSIS REQUIRED

Operation type: observation + estimate + comparison. **Interval/evidence gate:** Monthly and after a material near miss; require at least 90 valid days or 20 comparable events, stable threshold definitions and at least 85% time coverage for a trend statement. **Method:** Recurrence, rate-change and precursor clustering analysis. Deduplicate related alarms, calculate event rates and severity bands, apply control charts or Bayesian rate comparison, and stratify by season and operating mode. **Uncertainty/safe failure:** Publish rate ratios with credible or confidence intervals and missing-time adjustment; report 'no reliable trend' when event count, threshold stability or coverage is inadequate.

ANSWER BOUNDARY

The service must not predict a future loss, assign cause or negligence, identify people, change alarm thresholds, certify a control, or use a statistical association as a claims or coverage conclusion.

WHY THIS IS USEFUL

Helps trigger targeted inspection or maintenance assessment before recurring weak signals develop into a larger interruption or property loss.

IR-019

STAKEHOLDER QUESTION

Which hazards are introduced by contractors, deliveries, shared occupiers, common services or neighbouring works, and who can require and verify the controls?

SENSORS AND TELEMETRY

Proposed evidence: Permit and delivery-window status, temporary utility use, door and loading-bay cycles, dust, vibration, noise-free environmental indicators and declared isolation or barrier state at agreed interfaces. **Coverage/timing:** Mapped work, delivery and shared-service interfaces; event-level data during the authorised window, with pre-start and hand-back snapshots and no tracking of individuals. **Quality gate:** for every contributing stream, verify asset and location mapping, units, calibration or functional check, sampling and aggregation, clock offset, uptime, gaps, drift and source-to-result lineage; exclude any stream that fails its gate. Telemetry cannot observe safe working practice, cargo content or off-site activity; contractor records may be incomplete and neighbouring conditions may be outside permissioned coverage.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: Contract or tenant manager, project sponsor, permit issuer, estates, security, health and safety, procurement and the relevant third-party accountable owner. Contracts, leases, permits, method statements and competent site findings lead; owner-approved delivery and work records follow; telemetry only supports observed interface conditions and times. **Permissions/provenance:** Use company or role identifiers instead of personal tracking, retain permit and interface provenance, and leave contractual or factual conflicts visible for the relevant owner. **Time/staleness/conflicts:** retain contract or source effective, external observed and local retrieved times; expired or unverifiable third-party assertions are unknown; preserve differences with local records and identify the external and internal owners.

ANALYSIS REQUIRED

Operation type: lookup + comparison. **Interval/evidence gate:** Before higher-risk third-party activity and at each hand-back; require a current scope or permit, named control owner, mapped affected zones and at least 90% evidence coverage of the declared higher-risk period. **Method:** Third-party interface hazard and control-window assessment. Align work phases, access, temporary services, isolations and environmental indicators, then flag gaps between declared controls and observed interface states. **Uncertainty/safe failure:** Quantify unobserved time and unmapped interfaces; safe-fail to 'third-party exposure not established' where permission, scope, current owner or competent evidence is missing.

ANSWER BOUNDARY

The service must not monitor worker behaviour, infer cargo or activity content, assign contractual liability or negligence, approve work, enforce a permit, or decide coverage for third-party acts.

WHY THIS IS USEFUL

Supports a joint inspection, clearer evidence request, contract-owner escalation and proportionate controls around shared or temporary exposures.

IR-020

STAKEHOLDER QUESTION

Which missing, stale, conflicting, restricted or inaccessible evidence could materially alter the risk picture, and who owns each resolution?

SENSORS AND TELEMETRY

Proposed evidence: Source-inventory and health telemetry covering sensor mapping, uptime, calibration due dates, integration failures, unobserved zones and delayed records; this monitors evidence systems, not people. **Coverage/timing:** Every source used for a material statement; continuous health for live integrations and a purpose-specific freshness snapshot whenever the catalogue answer is generated. **Quality gate:** where telemetry is not applicable, validate identifiers, scope, status codes, signatures, timestamp and time-zone alignment, retrieval completeness, missing records, source availability, schema drift and transformation lineage; mark calibration and sampling not applicable. System health does not prove data correctness, accessible evidence may still be unauthoritative and restricted evidence may be legitimate but unavailable to this audience.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: Each authoritative source owner, data governance, estates, safety and security owners, competent assessors, policy or claims custodians and the requesting risk assessor. Policy terms, competent findings and current owner records lead within their scope; telemetry and analytics never fill an authoritative gap. **Permissions/provenance:** Record permission basis, provenance and reason for restriction, avoid copying restricted content, and present conflicts with named owners and affected conclusions. **Time/staleness/conflicts:** retain permission or source effective, evidence observed and query retrieved times; apply freshness to each fact and permission, not the file as a whole; preserve competing values and the accountable owner needed to resolve them.

ANALYSIS REQUIRED

Operation type: lookup + comparison. **Interval/evidence gate:** For every material answer; require a declared question, decision period, source inventory, owner, freshness rule and minimum-evidence gate before analysis begins. **Method:** Evidence-criticality, completeness and conclusion-sensitivity assessment. Score evidence by authority, coverage, age, integrity and relevance, then test whether each missing or conflicting item can reverse or widen the bounded conclusion. **Uncertainty/safe failure:** Report completeness percentages and conclusion sensitivity; use unknown, conflicting, restricted or unavailable rather than substituting assumptions when a material source fails.

ANSWER BOUNDARY

The service must not treat missing data as adverse behaviour, bypass permissions, disclose restricted records, invent values, silently reconcile authoritative conflicts, or make a consequential insurance decision.

WHY THIS IS USEFUL

Enables the assessor to distinguish a real risk concern from an evidence problem and request the smallest owner-led action needed to support a defensible view.

STAKEHOLDER QUESTION

IR-021 Which fire-detection, alarm, suppression and life-safety interfaces are available as declared now, and which exact zones are affected by any impairment?

SENSORS AND TELEMETRY

Proposed evidence: Fire-panel, loop, power-supply and alarm-transmission health; sprinkler valve, supply-pressure and pump status; and approved interface states for smoke control, door release or plant shutdown. **Coverage/timing:** All owner-declared fire systems and mapped protection zones; event capture within seconds, daily heartbeat summaries and explicit impairment start, acknowledgement, assessment and restoration times. **Quality gate:** verify point-to-asset or zone mapping and event-code semantics; test contacts, alarms and heartbeats; retain event aggregation, clock offset, uptime, lost events, configuration drift and source-to-summary lineage. Calibrate only underlying physical sensors. Panel status cannot prove detector sensitivity, water delivery, door operation or field audibility; unmapped devices, hidden valves and manual controls require competent inspection and testing.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: The responsible fire-safety owner, estates engineering, alarm-receiving or monitoring provider, competent maintainers and affected duty managers. Current competent test findings, fire strategy and signed impairment records lead within scope; owner registers follow; commissioned telemetry supports observed availability only. **Permissions/provenance:** Restrict detailed system topology and security-sensitive alarm information, preserve device-zone and event lineage, and escalate conflicting panel, test or impairment states to the competent fire owner. **Time/staleness/conflicts:** retain owner-record effective, event observed and query retrieved times; live status is stale after a missed heartbeat or 15 minutes unless the owner sets a stricter limit; preserve zone and time disagreements.

ANALYSIS REQUIRED

Operation type: lookup + observation + comparison. **Interval/evidence gate:** Continuous for critical states and before a risk survey; require current system and zone maps, latest competent tests, named impairment ownership and verified status for at least 95% of critical interfaces. **Method:** Fire-protection availability, zone-impact and impairment reconciliation. Align test, fault, isolation and restoration records, map each impairment to protected zones and identify simultaneous loss of independent layers or overdue temporary measures. **Uncertainty/safe failure:** Report evidenced device or interface coverage, unobserved zones and clock uncertainty; return 'protection state not established' where a critical mapping, test or current status is missing.

ANSWER BOUNDARY

The service must not certify fire safety or compliance, authorise occupation, direct evacuation or system operation, disclose restricted protection layouts, promise alarm response, or decide policy or claim treatment.

WHY THIS IS USEFUL

Enables immediate owner clarification, targeted specialist inspection, accurate impairment scoping and preservation of a bounded protection-state record.

STAKEHOLDER QUESTION

IR-022 Which fire doors, protected routes or emergency-access points show repeated open, failed-to-close or unavailable states that require competent inspection?

SENSORS AND TELEMETRY

Proposed evidence: Door-position and closer-cycle sensors on selected critical doors, hold-open release status, route or exit availability states and pressure or environmental cues where already approved; no cameras or identity data. **Coverage/timing:** Owner-prioritised fire doors, protected routes and emergency access points; event-level transitions with daily exception summaries and monthly physical validation samples. **Quality gate:** verify point-to-asset or zone mapping and event-code semantics; test contacts, alarms and heartbeats; retain event aggregation, clock offset, uptime, lost events, configuration drift and source-to-summary lineage. Calibrate only underlying physical sensors. Sensors cannot detect gaps, seals, hinges, damage, wedging, obstruction or fire resistance, and a closed contact may not mean the leaf is correctly latched.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: Fire-safety, estates, security, access and inclusive-evacuation owners, supported by competent fire-door inspectors. Current fire strategy, door register, competent inspection findings and approved route records lead; telemetry supports observed state patterns only. **Permissions/provenance:** Use coded door and route references, restrict sensitive access layouts, retain commissioning and validation lineage, and show register-versus-sensor conflicts for physical verification. **Time/staleness/conflicts:** retain owner-record effective, event observed and query retrieved times; live status is stale after a missed heartbeat or 15 minutes unless the owner sets a stricter limit; preserve zone and time disagreements.

ANALYSIS REQUIRED

Operation type: observation + comparison + diagnosis. **Interval/evidence gate:** Weekly for critical doors and before survey; require a current door register, verified mapping, at least 30 valid days and physical validation of a representative sample including every high-exception door. **Method:** Door-state exception, persistence and inspection-priority analysis. Calculate open-duration distributions, failed-close sequences and event clusters by operating period, then combine with the latest competent inspection age without replacing it. **Uncertainty/safe failure:** Report false-state rate from validation, observed coverage and exception confidence; safe-fail to 'condition unknown' where contact placement or door identity is uncertain.

ANSWER BOUNDARY

The service must not certify compartmentation or escape suitability, infer obstruction or misuse, identify users, authorise route closure, replace inspection, or state that a door is compliant or insured.

WHY THIS IS USEFUL

Helps direct a competent door or route inspection to persistent operational exceptions instead of relying on occasional observations alone.

IR-023

STAKEHOLDER QUESTION

Where and when do perimeter, door or access-control faults, forced-open events or outages create a material security-control exception?

SENSORS AND TELEMETRY

Proposed evidence: Door and window contacts, reader and controller health, request-to-exit status, alarm-zone events and aggregate access counts; identities, credentials, video and network content are excluded. **Coverage/timing:** Approved perimeter and high-value access zones; event-level capture with 15-minute and daily aggregates, separated into open, closed, event and maintenance periods. **Quality gate:** verify point-to-asset or zone mapping and event-code semantics; test contacts, alarms and heartbeats; retain event aggregation, clock offset, uptime, lost events, configuration drift and source-to-summary lineage. Calibrate only underlying physical sensors. Tailgating, legitimate propping, maintenance and contact bounce can resemble intrusion; unmonitored openings, camera blind spots and off-site threats are not assessed.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: Security operations, estates, local secure-area custodians, IT access-control support and incident-management owners. Current security design and owner-approved access arrangements lead, followed by verified incident and maintenance records; telemetry supports observed device events only. **Permissions/provenance:** Aggregate or pseudonymise access data, suppress small groups, restrict detailed layouts and credentials, retain threshold and device lineage, and present conflicting event classifications to security. **Time/staleness/conflicts:** retain permission effective and expiry, event observed and query retrieved times; operational status follows the owner's short freshness limit; preserve differences in minimised form without exposing restricted detail.

ANALYSIS REQUIRED

Operation type: observation + comparison. **Interval/evidence gate:** Weekly and after a material event or access change; require 60 valid days, at least 90% monitored-opening coverage for the stated perimeter and current maintenance and authorised-opening calendars. **Method:** Security-control availability and anomaly-rate analysis. Deduplicate device chatter, classify events by authorised operating state, compare rates and durations with matched periods, and validate sampled anomalies through security records. **Uncertainty/safe failure:** Report event-rate intervals, validation precision and unmonitored-opening share; do not claim an exposure change when event count or perimeter coverage is inadequate.

ANSWER BOUNDARY

The service must not identify or profile individuals, infer criminal intent or negligence, disclose restricted access arrangements, direct security response, certify protection, or make coverage or claims decisions.

WHY THIS IS USEFUL

Supports targeted security inspection, maintenance, temporary-control discussion and a proportionate request for restricted incident evidence through the correct channel.

IR-024

STAKEHOLDER QUESTION

Which water services, rooms and downstream assets have the greatest escape-of-water exposure, and is there qualified evidence of leakage or abnormal flow or pressure?

SENSORS AND TELEMETRY

Proposed evidence: Incoming and branch flow or pressure, submeter pulses, valve state, spot or cable leak sensors, sump level and local humidity near vulnerable assets or concealed pipe routes. **Coverage/timing:** Verified mains, risers, branches and nominated high-consequence zones; one-minute operational data, faster local event capture where supported, and owner-confirmed snapshots after valve or pipe changes. **Quality gate:** confirm pipe, drain or catchment mapping, units and range; retain calibration or known-state checks, sampling and aggregation, clock offset, uptime, gaps, fouling or zero drift and alarm lineage. Small leaks can fall below meter resolution, legitimate use can mask leakage, sensors cover only their local area and automatic shut-off state does not prove isolation effectiveness.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: Estates water and plumbing owners, utilities, maintenance, local asset custodians and incident-response owners. Current competent schematics, valve and maintenance records and confirmed incident findings lead; calibrated telemetry supports observed hydraulic or local wetness conditions only. **Permissions/provenance:** Restrict sensitive asset locations where needed, preserve meter, valve and zone lineage, document units and calibration, and expose drawing-versus-observation conflicts to the water-system owner. **Time/staleness/conflicts:** retain owner-record effective, event observed and query retrieved times; live status is stale after a missed heartbeat or 15 minutes unless the owner sets a stricter limit; preserve zone and time disagreements.

ANALYSIS REQUIRED

Operation type: observation + diagnosis + estimate. **Interval/evidence gate:** Continuous alerting under owner rules and monthly risk assessment; require verified meter topology, 14 stable comparable days, at least 95% interval coverage and local confirmation for any claimed leak location. **Method:** Water-balance, change-point and leak-localisation screening. Model expected flow by operating state, detect sustained residual or pressure anomalies, triangulate with local wetness and valve events, and exclude known flushing or maintenance windows. **Uncertainty/safe failure:** Report detection limit, estimated residual-flow range and candidate zones; return 'possible leak - location unconfirmed' when topology, resolution or local corroboration is insufficient.

ANSWER BOUNDARY

The service must not command valves, promise loss prevention, infer pipe condition or cause, certify water tightness, authorise re-entry, or decide liability, coverage or claim value.

WHY THIS IS USEFUL

Enables faster owner investigation, targeted leak survey, asset-protection action under existing procedures and preservation of a time-bounded event record.

STAKEHOLDER QUESTION

IR-025

Which roof drains, gullies, sumps and site low points show surcharge, blockage or water-level conditions that warrant drainage inspection?

SENSORS AND TELEMETRY

Proposed evidence: Rainfall intensity, roof or gully water level, sump level and pump state, drain differential indicators and water-on-floor sensors at owner-identified low points. **Coverage/timing:** Verified roof catchments, drainage routes, basements and vulnerable thresholds; one-minute data during rain, five-minute otherwise, with pre-storm health checks and post-event inspection timestamps. **Quality gate:** confirm pipe, drain or catchment mapping, units and range; retain calibration or known-state checks, sampling and aggregation, clock offset, uptime, gaps, fouling or zero drift and alarm lineage. Local gauges do not prove drainage capacity, blockage can occur between sensors and intense rainfall may be spatially variable; below-ground condition and waterproofing require competent survey.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: Estates drainage and roof owners, maintenance, project records owners, the relevant official weather or flood-data source owner and local asset custodians. Competent drainage surveys, as-built records and confirmed maintenance or incident findings lead; official event data leads for its measured location; site telemetry supports local observations only. **Permissions/provenance:** Retain catchment, sensor, unit, calibration and weather-source provenance, respect external licences, and show conflicts between drawings, inspections and observed drainage response. **Time/staleness/conflicts:** retain owner-record effective, event observed and query retrieved times; live status is stale after a missed heartbeat or 15 minutes unless the owner sets a stricter limit; preserve zone and time disagreements.

ANALYSIS REQUIRED

Operation type: observation + comparison + estimate. **Interval/evidence gate:** Before forecast severe weather and after significant events; require mapped drainage paths, verified low points, healthy sensors and at least 10 comparable rain events or a competent hydraulic basis. **Method:** Rainfall-response, storage-rise and pump-performance screening. Align rainfall intensity with level rise, pump cycles and recession time, compare event response with cleaned or known-clear periods, and flag unusually slow drainage or repeated overflow cues. **Uncertainty/safe failure:** Report event coverage, rainfall uncertainty and response bands; do not extrapolate a return period or capacity beyond observed or competently modelled conditions.

ANSWER BOUNDARY

The service must not certify flood resilience or waterproofing, predict exact flooding, control pumps, replace roof or drainage inspection, establish causation, or make insurance or claims decisions.

WHY THIS IS USEFUL

Helps prioritise drain cleaning, roof or basement survey, temporary protection and evidence collection before or after severe rainfall.

STAKEHOLDER QUESTION

IR-026

Which critical plant shows a load-adjusted change in vibration, temperature, current, pressure or cycling that warrants competent condition assessment?

SENSORS AND TELEMETRY

Proposed evidence: Appropriately mounted vibration and temperature sensors, motor current, run state, pressure, flow and cycle counts, with high-rate features calculated locally where raw waveforms are unnecessary. **Coverage/timing:** Owner-nominated critical pumps, fans, compressors, chillers and similar plant; seconds-to-minutes during operation, aggregated by run and load state with commissioning and maintenance epochs retained. **Quality gate:** confirm sensor mounting, asset identity, units and load context; retain calibration or functional checks, sample rate and aggregation, clock offset, uptime, gaps, mounting change, drift and feature lineage. Mounting, sensor orientation, variable load, control changes and short run periods can dominate trends; telemetry cannot reveal all failure modes or replace hands-on examination.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: Estates engineering, asset owner, maintenance provider, OEM or competent machinery specialist and business-continuity owner. OEM limits, competent condition findings and maintenance records lead; owner-approved operating envelopes follow; commissioned telemetry supports measured behaviour only. **Permissions/provenance:** Retain sensor mounting, units, feature extraction and model version, restrict proprietary service detail, and present conflicts between diagnostic, work-order and observed states. **Time/staleness/conflicts:** retain owner-record effective, event observed and query retrieved times; live status is stale after a missed heartbeat or 15 minutes unless the owner sets a stricter limit; preserve zone and time disagreements.

ANALYSIS REQUIRED

Operation type: observation + comparison + diagnosis. **Interval/evidence gate:** Weekly for critical plant and after maintenance; require at least 30 operating hours across three representative load bands, at least 85% run-time coverage and one validated healthy baseline epoch. **Method:** Load-stratified condition monitoring and anomaly screening. Compare robust features within matched load and mode, apply trend and multivariate distance tests, and check anomalies against maintenance, control and ambient changes. **Uncertainty/safe failure:** Report effect size, baseline range and false-alert rate from validation; safe-fail to 'abnormal behaviour - diagnosis required' rather than naming a defect.

ANSWER BOUNDARY

The service must not diagnose failure, estimate remaining life without a competent method, override controls, instruct maintenance, certify machinery safety, or decide insurance or claim consequences.

WHY THIS IS USEFUL

Supports earlier competent inspection, spare or contingency planning and a focused maintenance evidence request for genuinely changing plant behaviour.

IR-027

STAKEHOLDER QUESTION

Which electrical distribution, UPS, battery or standby-power conditions show a qualified exception relevant to fire, equipment damage or interruption?

SENSORS AND TELEMETRY

Proposed evidence: Phase current and voltage, power quality, breaker trip or alarm state, cabinet temperature, battery voltage, current and temperature, UPS load and runtime estimate, and generator start or availability status. **Coverage/timing:** Owner-designated critical boards, UPS units, batteries and standby supplies; sub-minute event capture, five-minute trends and snapshots during tests, transfers and abnormal demand. **Quality gate:** confirm meter, CT, phase and load mapping plus units and sign convention; retain calibration class or functional check, interval aggregation, clock offset, uptime, gaps, configuration drift and meter-to-summary lineage. Fixed telemetry cannot see loose connections, insulation defects or all local heating, battery state-of-health estimates depend on model assumptions and load changes can mimic deterioration.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: Estates electrical owner, competent electrical inspectors, UPS or generator maintainer, IT or critical-load owners, fire-safety and business-continuity leads. Competent inspection and test findings, protection studies, OEM requirements and approved load schedules lead; telemetry supports observed operating conditions only. **Permissions/provenance:** Restrict detailed critical-power topology, preserve board, circuit, unit, calibration and test lineage, and show conflicts between labels, drawings, alarms and observed load. **Time/staleness/conflicts:** retain owner-record effective, event observed and query retrieved times; live status is stale after a missed heartbeat or 15 minutes unless the owner sets a stricter limit; preserve zone and time disagreements.

ANALYSIS REQUIRED

Operation type: observation + comparison + diagnosis. **Interval/evidence gate:** Weekly and after trips, tests or load change; require current inspection evidence, verified mapping, four representative weeks and at least 95% interval coverage for trend claims. **Method:** Electrical operating-envelope, imbalance and backup-readiness screening. Calculate load factor, imbalance, voltage and temperature excursions, battery trend and transfer outcomes, stratified by operating mode and ambient condition. **Uncertainty/safe failure:** Report instrument uncertainty, unexplained load share and confidence by asset; stop and escalate urgent owner-rule breaches, otherwise return 'specialist assessment required' without a diagnosis.

ANSWER BOUNDARY

The service must not certify electrical or fire safety, infer concealed defects, open equipment, command transfer or shutdown, expose restricted topology, or make underwriting, coverage or claim decisions.

WHY THIS IS USEFUL

Helps target competent electrical inspection, load assessment, battery testing and continuity planning before a trip, fire or prolonged outage.

IR-028

STAKEHOLDER QUESTION

Where are temperature, humidity, particulates or surface-dew-point conditions outside the owner-specified envelope for declared property or critical equipment?

SENSORS AND TELEMETRY

Proposed evidence: Zone and enclosure temperature and relative humidity, surface or dew-point proxies, particulate or dust sensing where relevant, leak status and environmental-control run state; no health inference is made. **Coverage/timing:** Owner-nominated sensitive assets, stores and supporting plant zones; one- to five-minute readings aggregated hourly, with separate occupied, unoccupied and maintenance modes. **Quality gate:** confirm representative siting, asset or zone mapping, units, traceable calibration or functional check, sampling and aggregation, clock offset, uptime, gaps, response lag, drift and raw-to-summary lineage. Room sensors may not represent equipment interiors, particulate sensors are composition-blind and calibration drift or solar gain can distort readings; health effects and unmonitored microclimates are outside scope.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: Asset or collections custodians, estates environmental-control owner, local activity owner, maintenance and competent conservation or equipment specialists. Owner-approved environmental requirements and competent specialist findings lead; BMS set-points are operational records, not requirements; telemetry supports measured conditions only. **Permissions/provenance:** Use coded sensitive locations, retain units, calibration, siting and aggregation lineage, and show conflicts between requirements, set-points and observed conditions to the relevant owner. **Time/staleness/conflicts:** retain owner-record effective, event observed and query retrieved times; live status is stale after a missed heartbeat or 15 minutes unless the owner sets a stricter limit; preserve zone and time disagreements.

ANALYSIS REQUIRED

Operation type: observation + comparison. **Interval/evidence gate:** Daily for critical assets and monthly assessment; require a current owner-approved envelope, verified representative placement, 14 valid days and at least 95% interval completeness. **Method:** Environmental-envelope, excursion-duration and condensation-risk screening. Calculate degree-hours outside limits, dew-point margin and concurrent control state, compare with matched season and operating mode, and identify sustained rather than momentary exceptions. **Uncertainty/safe failure:** Report measurement uncertainty, representativeness limits and excursion ranges; safe-fail to unknown when no authoritative requirement or representative sensor exists.

ANSWER BOUNDARY

The service must not infer personal health or comfort, certify preservation or environmental compliance, control HVAC, disclose restricted asset locations, attribute damage cause, or decide claim response.

WHY THIS IS USEFUL

Supports a specialist inspection, environmental-control assessment, temporary protection and preservation of evidence around a material excursion.

IR-029

STAKEHOLDER QUESTION

Which lifts, automatic doors or other moving building systems show abnormal faults, cycles or unavailability that may increase damage or interruption exposure?

SENSORS AND TELEMETRY

Proposed evidence: Controller fault and availability state, cycle counts, door-reopen or obstruction events, motor current and temperature where supported, emergency-power state and maintenance-test timestamps. **Coverage/timing:** Owner-designated lifts, automatic doors, shutters and similar systems; event capture within seconds, daily availability summaries and operating-cycle aggregation without user identities. **Quality gate:** verify point-to-asset or zone mapping and event-code semantics; test contacts, alarms and heartbeats; retain event aggregation, clock offset, uptime, lost events, configuration drift and source-to-summary lineage. Calibrate only underlying physical sensors. Controller codes may be vendor-specific, high cycles can reflect legitimate demand and telemetry cannot assess mechanical wear, safety devices or passenger experience.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: Estates vertical-transport or access-system owner, competent examiner, maintenance contractor, accessibility owner, security and business-continuity leads. Competent examination and maintenance findings, OEM records and owner-approved service status lead; telemetry supports observed faults, cycles and downtime only. **Permissions/provenance:** Preserve asset, code, firmware, mapping and maintenance lineage, restrict proprietary diagnostics and show conflicts between controller, ticket and contractor records. **Time/staleness/conflicts:** retain owner-record effective, event observed and query retrieved times; live status is stale after a missed heartbeat or 15 minutes unless the owner sets a stricter limit; preserve zone and time disagreements.

ANALYSIS REQUIRED

Operation type: observation + comparison + estimate. **Interval/evidence gate:** Monthly and after a serious fault; require 90 valid days or 500 cycles, at least 90% event capture, current examination evidence and known maintenance windows. **Method:** Fault-rate, cycle-normalised reliability and downtime analysis. Deduplicate linked faults, calculate failures per cycle, remedial-work and return-to-service intervals, compare matched operating periods and identify repeat fault families. **Uncertainty/safe failure:** Report rate intervals, unobserved downtime and vendor-code uncertainty; use 'abnormal pattern - competent assessment required' rather than declaring unsafe condition.

ANSWER BOUNDARY

The service must not certify statutory safety, diagnose mechanical failure, dispatch or control equipment, identify users, authorise return to service, or decide insurance or claim outcomes.

WHY THIS IS USEFUL

Enables targeted competent examination, maintenance challenge, accessibility contingency planning and accurate interruption evidence.

IR-030

STAKEHOLDER QUESTION

Which physically credible combinations of fire, water, power, security, machinery or environmental events could propagate beyond one zone or service?

SENSORS AND TELEMETRY

Proposed evidence: Time-aligned state and event data from the relevant fire, water, electrical, environmental, security and plant systems, plus verified dependency and zone mappings. **Coverage/timing:** Owner-selected high-consequence assets and cross-system dependencies; seconds-to-minutes for event sequence, with configuration snapshots for each modelled scenario period. **Quality gate:** for every contributing stream, verify asset and location mapping, units, calibration or functional check, sampling and aggregation, clock offset, uptime, gaps, drift and source-to-result lineage; exclude any stream that fails its gate. Multi-system clocks, event semantics and coverage differ, rare combinations may never have occurred and correlation cannot establish causal propagation.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: Estates engineering, fire, security, utilities, IT, asset custodians, incident management and business-continuity owners, supported by competent specialists. Competent hazard studies, fire strategy, engineering schematics and current owner dependency records lead; incidents and telemetry support observed sequences only. **Permissions/provenance:** Restrict sensitive topology, retain event, mapping and model lineage, and display every assumption or source conflict rather than creating a seamless synthetic history. **Time/staleness/conflicts:** retain topology effective or version, supporting observed and query retrieved times; a topology becomes stale after material change; show drawing, register and telemetry differences without selecting a winner.

ANALYSIS REQUIRED

Operation type: calculation + estimate + comparison. **Interval/evidence gate:** At annual risk assessment and after material change; require validated mappings for each scenario path, clock alignment within the claimed sequence tolerance and evidence for at least 90% of critical nodes. **Method:** Event-tree and dependency-graph cascade scenario analysis. Define initiating events and conditional transitions with competent owners, test single and selected concurrent failures, and compare modelled paths with drills or incidents where available. **Uncertainty/safe failure:** Present outcome ranges and sensitivity to uncertain transitions, not a single expected loss; return 'scenario not supportable' when a critical dependency or causal step lacks evidence.

ANSWER BOUNDARY

The service must not calculate a probable maximum loss, predict an incident, assign causation or liability, disclose restricted system architecture, control protective systems, or make underwriting decisions.

WHY THIS IS USEFUL

Helps scope specialist surveys, challenge common-cause assumptions and prioritise evidence or resilience actions around plausible cascading loss paths.

IR-031

STAKEHOLDER QUESTION

What is the best-supported incident timeline from the last known normal state through detection, response, containment and restoration, and where are the unresolved gaps?

SENSORS AND TELEMETRY

Proposed evidence: Unaltered event and state logs from relevant alarms, BMS, water, power, plant, access and communications systems, plus approved manual observation timestamps; identities are excluded unless separately authorised and necessary. **Coverage/timing:** Only systems and zones within the incident scope; preserve native event resolution from the earliest relevant pre-event window through owner-confirmed restoration, without resampling the evidential copy. **Quality gate:** verify point-to-asset or zone mapping and event-code semantics; test contacts, alarms and heartbeats; retain event aggregation, clock offset, uptime, lost events, configuration drift and source-to-summary lineage. Calibrate only underlying physical sensors. Clocks may drift, logs may overwrite, communications can arrive late and a recorded state does not prove a physical action; missing systems and manual activity create sequence intervals rather than exact times.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: Incident command or response owner, each system custodian, security, estates, evidence or claims-record custodian and any appointed competent investigator. Signed incident logs and competent investigative findings lead within their scope; native system logs support recorded events. **Permissions/provenance:** Capture read-only exports with hashes, source and clock metadata, permission basis and transformation lineage; keep conflicting timestamps visible and route them to the evidence owner. **Time/staleness/conflicts:** retain record effective or version, event observed and extraction or retrieved times; freeze each extract; later amendments keep original and amended timestamps, and conflicting accounts remain separate until the incident owner resolves scope.

ANALYSIS REQUIRED

Operation type: observation + calculation + comparison. **Interval/evidence gate:** As soon as authorised after an incident; require preserved native logs, known clock status, at least two independent sources for each material transition and documented gaps. **Method:** Forensic timeline reconstruction with clock-offset correction. Normalise time zones, estimate clock offsets from shared events, order events as exact times or bounded intervals, and separate observed fact, owner statement and analytic inference. **Uncertainty/safe failure:** Publish timing tolerances and alternative sequences where ordering is unresolved; safe-fail to 'sequence indeterminate' rather than filling missing events or choosing a convenient source.

ANSWER BOUNDARY

The service must not determine cause, fault, liability, policy response or claim value, infer unrecorded human actions, alter source evidence, disclose restricted logs, or replace a competent investigation.

WHY THIS IS USEFUL

Provides a traceable chronology for specialist investigation, evidence preservation, response assessment and bounded claims fact-finding.

IR-032

STAKEHOLDER QUESTION

For each plausible incident-cause hypothesis, which observations support it, contradict it or only reflect downstream consequences?

SENSORS AND TELEMETRY

Proposed evidence: Preserved pre-event and event-period condition data relevant to the hypotheses, such as flow, pressure, temperature, electrical, vibration, door, alarm and environmental states, tied to verified assets and zones. **Coverage/timing:** The competent investigator's defined incident scope and a pre-event baseline long enough for comparison; native event resolution retained and post-event data separated from pre-event evidence. **Quality gate:** for every contributing stream, verify asset and location mapping, units, calibration or functional check, sampling and aggregation, clock offset, uptime, gaps, drift and source-to-result lineage; exclude any stream that fails its gate. Sensors may fail during the event, common effects can appear across several causes and absence of a recorded anomaly is weak evidence when coverage or detection limits are poor.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: The appointed competent investigator, incident owner, relevant engineering or safety specialists, system custodians and evidence or claims-record custodian. Competent investigative findings and verified physical evidence lead; native records support observed conditions only for every item. **Permissions/provenance:** Preserve chain of custody, hypothesis version, source lineage and contradictory evidence; restrict legally privileged, security-sensitive or personal material to authorised routes. **Time/staleness/conflicts:** retain record effective or version, event observed and extraction or retrieved times; freeze each extract; later amendments keep original and amended timestamps, and conflicting accounts remain separate until the incident owner resolves scope.

ANALYSIS REQUIRED

Operation type: diagnosis + comparison. **Interval/evidence gate:** Only after the investigator defines plausible hypotheses; require at least one predicted observation for and against each hypothesis, verified source coverage and explicit alternative explanations. **Method:** Hypothesis-evidence matrix and bounded causal screening. Compare temporal order, physical compatibility and expected signatures, use likelihood ratios or qualitative support bands where justified, and test sensitivity to missing sources. **Uncertainty/safe failure:** State evidence support, contradiction and coverage separately; do not assign a probability when priors or detection performance are unknown, and return 'cause not established' when hypotheses remain indistinguishable.

ANSWER BOUNDARY

The service must not declare proximate cause, liability, negligence, fraud, policy coverage or claim entitlement, infer personal intent, or override the competent investigator's scope and findings.

WHY THIS IS USEFUL

Helps the investigator and assessor request discriminating evidence and avoid confusing correlation, response effects and damage consequences with cause.

IR-033

STAKEHOLDER QUESTION

Which declared assets, spaces and services were physically or operationally affected, for how long, and what extent remains unverified?

SENSORS AND TELEMETRY

Proposed evidence: Asset and zone status, environmental excursions, power or service availability, leak or alarm spread, work-order state and owner-approved access or closure indicators, without tracking individuals. **Coverage/timing:** All incident-scoped zones and owner-declared assets from the first impact through stable restoration; one- to fifteen-minute operational data plus owner-confirmed inspection and hand-back times. **Quality gate:** for every contributing stream, verify asset and location mapping, units, calibration or functional check, sampling and aggregation, clock offset, uptime, gaps, drift and source-to-result lineage; exclude any stream that fails its gate. Telemetry may remain normal despite hidden damage, mobile or untaged assets can be missed and closure does not prove physical damage; inaccessible areas remain unknown until surveyed.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: Incident owner, estates, space management, asset custodians, finance registers, business-continuity owners and appointed loss or technical assessors. Competent physical findings and signed owner damage or closure records lead; current asset registers define declared scope; telemetry supports observed service and environmental effects only. **Permissions/provenance:** Use coded assets and zones where restricted, preserve source/version and incident lineage, and expose conflicts between asset, inspection, closure and telemetry records. **Time/staleness/conflicts:** retain record effective or version, event observed and extraction or retrieved times; freeze each extract; later amendments keep original and amended timestamps, and conflicting accounts remain separate until the incident owner resolves scope.

ANALYSIS REQUIRED

Operation type: lookup + observation + calculation. **Interval/evidence gate:** After initial containment and at each recovery milestone; require a current scoped asset list, owner-confirmed zone access state and evidence covering at least 90% of nominated critical services. **Method:** Spatial-temporal impact inventory and restoration-state classification. Join assets to affected zones and dependencies, classify confirmed, indicated and unassessed impact, and calculate outage or excursion duration with partial-restoration states. **Uncertainty/safe failure:** Report affected-count and duration ranges, inaccessible share and confidence by zone; do not convert operational interruption into physical damage or monetary loss without competent evidence.

ANSWER BOUNDARY

The service must not value damage or business interruption, determine claim scope, coverage or causation, disclose restricted assets, authorise reoccupation, or treat a service alarm as proof of damage.

WHY THIS IS USEFUL

Supports targeted inspection, evidence requests, recovery tracking and a transparent statement of what is confirmed versus still unassessed.

IR-034

STAKEHOLDER QUESTION

Can every digital evidence item be authenticated, scoped, permissioned and preserved with an unbroken chain from collection to analysis?

SENSORS AND TELEMETRY

Proposed evidence: Source-health, system-event-log, export, checksum, signature and retention-status evidence for the sensors, gateways, databases and analytic extracts used; no new content collection beyond the authorised incident scope. **Coverage/timing:** Every digital item relied on for a material statement; native resolution and original format preserved, with acquisition, transfer, transformation and access events timestamped. **Quality gate:** where telemetry is not applicable, validate identifiers, scope, status codes, signatures, timestamp and time-zone alignment, retrieval completeness, missing records, source availability, schema drift and transformation lineage; mark calibration and sampling not applicable. A valid hash proves file consistency after hashing, not truth at source; missing assurance logs, mutable databases, undocumented filters and clock uncertainty reduce evidential weight.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: The source-system custodian, information governance, security, incident or claims evidence custodian, legal or records owner and appointed investigator. Formal evidence-handling procedures and source-system records lead; native data and cryptographic checks support integrity claims only within the documented chain. **Permissions/provenance:** Record lawful or contractual purpose, requester, access, source identifier, hash, export method and every transformation; do not copy restricted evidence into wider working sets. **Time/staleness/conflicts:** retain permission or source effective, evidence observed and query retrieved times; apply freshness to each fact and permission, not the file as a whole; preserve competing values and the accountable owner needed to resolve them.

ANALYSIS REQUIRED

Operation type: lookup + observation + comparison. **Interval/evidence gate:** Before analytic use and at every hand-off; require an immutable evidential copy, source and acquisition metadata, hash verification, permission record and a stated scope-completeness test. **Method:** Chain-of-custody, integrity and completeness verification. Validate hashes, access and transformation checks, compare sequence counts and retention windows, and reproduce derived outputs from the preserved source where permitted. **Uncertainty/safe failure:** State integrity, completeness and permission as separate statuses; safe-fail to restricted, incomplete or unverifiable where any material chain element is missing.

ANSWER BOUNDARY

The service must not claim legal admissibility, authenticity beyond the documented chain, disclose restricted evidence, alter originals, infer missing records, or decide fraud, liability, coverage or claims outcomes.

WHY THIS IS USEFUL

This evidence protects evidential value, reveals where conclusions rely on incomplete records and gives investigators a reproducible path from source to statement.

IR-035

STAKEHOLDER QUESTION

Are near misses, minor losses or control activations changing in frequency, severity, location or sequence, and is the apparent trend statistically credible?

SENSORS AND TELEMETRY

Proposed evidence: Owner-classified near-miss and incident records linked to relevant alarm, leak, fault, environmental or work-order events, aggregated by asset, zone and operating period rather than person. **Coverage/timing:** All comparable reported events within the defined hazard scope; event-level capture with monthly rates and an exposure denominator such as operating hours, occupancy band or asset count. **Quality gate:** where telemetry is not applicable, validate identifiers, scope, status codes, signatures, timestamp and time-zone alignment, retrieval completeness, missing records, source availability, schema drift and transformation lineage; mark calibration and sampling not applicable. Reporting practice can change, duplicate alarms can inflate counts and severe events are sparse; unreported near misses cannot be inferred from telemetry alone.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: Incident and near-miss owner, health and safety, estates, security, fire, local custodians and the evidence governance owner. Confirmed incident and near-miss classifications lead; maintenance and telemetry records support event context only. **Permissions/provenance:** Use zone and asset identifiers, minimise personal details, retain classification and deduplication lineage, and show changed reporting definitions or conflicting event classes. **Time/staleness/conflicts:** retain permission or source effective, evidence observed and query retrieved times; apply freshness to each fact and permission, not the file as a whole; preserve competing values and the accountable owner needed to resolve them.

ANALYSIS REQUIRED

Operation type: observation + estimate + comparison. **Interval/evidence gate:** Quarterly and after a serious event; require at least 12 comparable months or 30 events, stable classification rules, a defensible exposure denominator and at least 85% relevant telemetry coverage. **Method:** Exposure-adjusted trend, clustering and precursor-sequence analysis. Deduplicate linked records, calculate rates and control-chart limits, test spatial or temporal clustering and compare precursor occurrence before versus outside serious-event windows. **Uncertainty/safe failure:** Publish rate intervals, reporting-change sensitivity and event-count limits; return 'trend not estimable' rather than treating no reports as no near misses.

ANSWER BOUNDARY

The service must not predict a future loss, profile individuals, infer unsafe behaviour or negligence, suppress reporting, establish causation, or use trends to decide policy or claim treatment.

WHY THIS IS USEFUL

Helps focus preventative inspection, control assessment and evidence collection on recurring weak signals rather than waiting for a major loss.

IR-036

STAKEHOLDER QUESTION

Do separate incidents share an evidenced asset type, control, contractor, location, environmental state or upstream dependency?

SENSORS AND TELEMETRY

Proposed evidence: Normalised incident metadata and linked condition, maintenance, contractor, weather and dependency records, using coded entities and no person-level behavioural analysis. **Coverage/timing:** All owner-approved comparable incidents in the chosen period and scope; preserve original event dates and configurations, with derived features calculated consistently across cases. **Quality gate:** where telemetry is not applicable, validate identifiers, scope, status codes, signatures, timestamp and time-zone alignment, retrieval completeness, missing records, source availability, schema drift and transformation lineage; mark calibration and sampling not applicable. Small samples create unstable clusters, shared labels may reflect reporting habits and common context does not establish common cause; older configurations may no longer be comparable.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: Incident management, estates, security, fire and safety owners, procurement or contractor managers, asset custodians and competent investigators. Competent findings and confirmed incident classifications lead; current owner records define assets and dependencies for each historical date; telemetry supports observed context only. **Permissions/provenance:** Pseudonymise restricted parties where possible, retain feature and linkage provenance, and show uncertain entity matches or conflicting classifications instead of forcing a common identity. **Time/staleness/conflicts:** retain permission or source effective, evidence observed and query retrieved times; apply freshness to each fact and permission, not the file as a whole; preserve competing values and the accountable owner needed to resolve them.

ANALYSIS REQUIRED

Operation type: lookup + comparison + estimate. **Interval/evidence gate:** Annually and after repeated material events; require at least five comparable incidents or complete assessment of all material incidents, stable definitions and historical configuration evidence for each case. **Method:** Cross-incident similarity, association and dependency analysis. Build an incident-feature matrix, apply exact matching and interpretable clustering, calculate association strength, and test whether findings persist when one case or feature is removed. **Uncertainty/safe failure:** Report cluster stability, sample size and alternative groupings; use 'possible common factor' rather than common cause unless a competent investigation establishes it.

ANSWER BOUNDARY

The service must not identify or blame individuals, contractors or occupiers, infer negligence or fraud, declare causation, disclose restricted incident details, or decide liability, coverage or claims handling.

WHY THIS IS USEFUL

Supports a portfolio-style specialist assessment, targeted control inspection and coordinated corrective action where repeated events may share an underlying dependency.

IR-037

STAKEHOLDER QUESTION

Which incident and loss evidence can each authorised party receive for this purpose, and what needs minimisation, redaction or controlled access?

SENSORS AND TELEMETRY

Proposed evidence: Permission, classification, retention and access-control metadata for evidence repositories; sensors are not queried until the purpose, audience, fields and time window are approved. **Coverage/timing:** Every evidence item proposed for sharing; evaluate at request time because permissions, investigations, legal holds and sensitivity can change. **Quality gate:** where telemetry is not applicable, validate identifiers, scope, status codes, signatures, timestamp and time-zone alignment, retrieval completeness, missing records, source availability, schema drift and transformation lineage; mark calibration and sampling not applicable. Technical access does not equal permission, redaction can remove context and aggregate data can still reveal sensitive locations or small groups.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: Information governance, legal or records owner, incident commander, security, source-system custodians, claims or policy custodian and the requesting party's accountable sponsor. Law, policy terms, legal holds, investigation instructions and source-owner classifications lead; prior access never creates authority for a new purpose. **Permissions/provenance:** Apply least privilege, purpose limitation, field and row minimisation, secure delivery and full access logging; record withheld categories and conflicts without exposing their content. **Time/staleness/conflicts:** retain permission effective and expiry, event observed and query retrieved times; operational status follows the owner's short freshness limit; preserve differences in minimised form without exposing restricted detail.

ANALYSIS REQUIRED

Operation type: lookup + comparison + recommendation. **Interval/evidence gate:** For every disclosure; require a named recipient role, stated purpose, legal or contractual basis, owner approval, necessary fields, retention period and secure transfer route. **Method:** Purpose-to-permission matrix and disclosure minimisation assessment. Map requested statements to the minimum source fields, test re-identification or security exposure, apply approved redaction or aggregation, and verify the delivered package against the decision record. **Uncertainty/safe failure:** State permitted, restricted and unavailable evidence separately; safe-fail to no disclosure when authority, purpose or secure route is unresolved.

ANSWER BOUNDARY

The service must not bypass access controls, expose personal, security-sensitive, privileged or commercially restricted records, infer withheld content, or decide entitlement to policy or claim information.

WHY THIS IS USEFUL

This evidence lets assessors obtain the evidence genuinely needed while protecting investigations, people, security controls and contractual confidentiality.

IR-038

STAKEHOLDER QUESTION

For how long were critical spaces or services unavailable, degraded or under workaround, and what evidence marks partial and full restoration?

SENSORS AND TELEMETRY

Proposed evidence: Service availability, environmental state, power, network, plant, access and alarm status linked to owner-approved closure, workaround, test and hand-back records. **Coverage/timing:** Incident-scoped critical services and zones from first confirmed impact to owner-accepted stable operation; retain one- to fifteen-minute state and exact milestone timestamps. **Quality gate:** for every contributing stream, verify asset and location mapping, units, calibration or functional check, sampling and aggregation, clock offset, uptime, gaps, drift and source-to-result lineage; exclude any stream that fails its gate. A service can be technically available but operationally unusable, temporary workaround may have reduced capacity and return-to-service alarms do not prove accepted restoration.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: Incident and recovery owner, estates, IT, security, space management, local service owners, business continuity and competent technical specialists. Signed closure, workaround and hand-back decisions lead; competent test findings support technical restoration; telemetry supports observed service state only. **Permissions/provenance:** Use role and service identifiers rather than personal data, retain milestone and state lineage, and show differences between technical availability, owner acceptance and user-service restoration. **Time/staleness/conflicts:** retain record effective or version, event observed and extraction or retrieved times; freeze each extract; later amendments keep original and amended timestamps, and conflicting accounts remain separate until the incident owner resolves scope.

ANALYSIS REQUIRED

Operation type: observation + calculation + comparison. **Interval/evidence gate:** During recovery and at close-out; require defined critical services, owner-approved state definitions, incident start basis and evidence for at least 90% of the claimed interruption period. **Method:** Multi-state interruption duration and recovery-milestone analysis. Construct a state timeline for unavailable, degraded, workaround, tested and accepted operation, calculate durations and identify unsupported or overlapping transitions. **Uncertainty/safe failure:** Report lower and upper duration bounds when start or restoration is uncertain; do not collapse partial service into full availability or fill unobserved intervals.

ANSWER BOUNDARY

The service must not calculate insured business interruption, value loss, certify safe return, authorise occupation, promise recovery, or decide claim period, coverage or settlement.

WHY THIS IS USEFUL

Provides a defensible operational chronology for continuity assessment, specialist assessment and claims fact-finding without turning service data into a monetary decision.

STAKEHOLDER QUESTION

IR-039 Which third parties or external platforms may hold material incident evidence, and what exact records, identifiers and time windows should be requested?

SENSORS AND TELEMETRY

Proposed evidence: Interface event metadata such as utility outage state, alarm-receiving timestamps, contractor attendance or service logs, delivery or access windows and official weather observations, subject to owner permission. **Coverage/timing:** Only verified external dependencies and incident-relevant interfaces; define a bounded pre-event, event and recovery window before requesting data. **Quality gate:** where telemetry is not applicable, validate identifiers, scope, status codes, signatures, timestamp and time-zone alignment, retrieval completeness, missing records, source availability, schema drift and transformation lineage; mark calibration and sampling not applicable. External logs can use different clocks and retention periods, licences may restrict reuse and the building service cannot verify records it has not received.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: Utility, monitoring, maintenance, contractor, landlord, tenant, weather or emergency-service record owners, coordinated through the incident, legal or claims evidence custodian. The external owner's native record and competent findings lead within their scope; internal interface logs provide corroboration only. **Permissions/provenance:** Use formal authorised routes, request only necessary fields, retain request and response provenance, and document unavailable, refused or conflicting external evidence without speculation. **Time/staleness/conflicts:** retain contract or source effective, external observed and local retrieved times; expired or unverifiable third-party assertions are unknown; preserve differences with local records and identify the external and internal owners.

ANALYSIS REQUIRED

Operation type: lookup + recommendation. **Interval/evidence gate:** As early as possible after an incident; require a verified dependency or involvement basis, named external owner, exact fields, event window, retention urgency and authorised requester. **Method:** External-evidence source mapping and time-window specification. Map each unanswered material timeline or cause question to the external source most capable of resolving it, account for clock offsets and prioritise evidence at risk of deletion. **Uncertainty/safe failure:** State expected evidential value and retrieval uncertainty; safe-fail to 'external evidence not obtained' rather than inferring its content or completeness.

ANSWER BOUNDARY

The service must not compel disclosure, interpret legal duties, infer third-party fault, contact parties outside authorised routes, expose restricted information, or decide liability, coverage or claims outcomes.

WHY THIS IS USEFUL

Helps preserve short-retention external records and makes evidence requests precise enough to support investigation without unnecessary disclosure.

STAKEHOLDER QUESTION

IR-040 Were post-incident corrective actions completed and accepted as specified, and has the same precursor, control failure or loss mechanism reappeared?

SENSORS AND TELEMETRY

Proposed evidence: Action and work-order state, completion evidence references, linked control status and comparable post-remediation condition or event indicators, without worker surveillance. **Coverage/timing:** Every agreed material corrective action and its affected asset, zone or activity; status at each due date, completion and a defined follow-up period appropriate to the failure mode. **Quality gate:** verify point-to-asset or zone mapping and event-code semantics; test contacts, alarms and heartbeats; retain event aggregation, clock offset, uptime, lost events, configuration drift and source-to-summary lineage. Calibrate only underlying physical sensors. Ticket closure can precede effective remedial work, changed operation can suppress recurrence and absence of an event in a short window is weak evidence of effectiveness.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: Incident action owner, competent investigator or specialist, estates, security, fire or safety control owner, contractor manager and local asset custodian. Signed incident recommendations and competent acceptance criteria lead; owner-approved completion and test evidence follow; telemetry supports observed post-action behaviour only. **Permissions/provenance:** Retain action-to-finding, work, test and monitoring lineage, restrict personnel and commercial detail, and show inconsistent closure, test or recurrence classifications. **Time/staleness/conflicts:** retain action effective or version, linked observed and query retrieved times; status without a dated owner update or acceptance evidence is stale; keep open and closed records unresolved until the accountable owner confirms the state.

ANALYSIS REQUIRED

Operation type: lookup + observation + comparison. **Interval/evidence gate:** At each due date and incident close-out; require stated acceptance criteria, competent completion evidence and a follow-up window containing at least 90% valid relevant observations or a formal test. **Method:** Corrective-action closure and recurrence-effectiveness assessment. Verify evidence against each acceptance criterion, compare pre- and post-action event rates or condition distributions, and test whether any recurrence has the same mapped mechanism. **Uncertainty/safe failure:** Report closure evidence completeness and post-action exposure time; use 'effectiveness not yet demonstrated' when follow-up is too short, conditions differ or evidence is missing.

ANSWER BOUNDARY

The service must not certify remedial-work quality, declare causation resolved, infer individual or contractor negligence, close an accountable action, or decide claim recovery, coverage or underwriting consequences.

WHY THIS IS USEFUL

Supports accountable close-out, targeted reinspection and escalation where a supposedly treated precursor or failure mode remains evident.

STAKEHOLDER QUESTION

IR-041 Which loss-control inspections, competent examinations and functional tests are due, overdue, incomplete or unsupported for the assets they claim to cover?

SENSORS AND TELEMETRY

Proposed evidence: Approved maintenance-schedule feeds, device self-test or test-mode records, inspection check-in timestamps and evidence-link health; telemetry does not replace the physical or functional test. **Coverage/timing:** All owner-designated safety-critical and property-protection controls; daily due-status snapshots and event timestamps for appointment, attendance, test, defect and certificate issue. **Quality gate:** where telemetry is not applicable, validate identifiers, scope, status codes, signatures, timestamp and time-zone alignment, retrieval completeness, missing records, source availability, schema drift and transformation lineage; mark calibration and sampling not applicable. Calendar entries can be wrong, self-tests cover limited functions and an uploaded document may not match the asset or scope; inaccessible and unregistered assets remain outside the schedule.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: The competent person or test provider, estates compliance and maintenance owners, fire, security, water, electrical and machinery control owners and records governance. Applicable owner-approved inspection regimes and competent reports lead; maintenance schedules support due dates and telemetry only corroborates test events. **Permissions/provenance:** Retain asset, standard or regime, scope, provider, result and evidence-link provenance, restrict sensitive findings and show conflicting intervals, assets or completion states. **Time/staleness/conflicts:** retain permission or source effective, evidence observed and query retrieved times; apply freshness to each fact and permission, not the file as a whole; preserve competing values and the accountable owner needed to resolve them.

ANALYSIS REQUIRED

Operation type: lookup + calculation + comparison. **Interval/evidence gate:** Weekly for critical controls and before risk assessment; require a current asset population, owner-approved interval, last valid evidence, next due date and competent owner for at least 98% of material controls. **Method:** Inspection-schedule completeness, scope and overdue analysis. Join assets to regimes and evidence, distinguish scheduled, attended, completed, passed and closed states, and calculate overdue days and unsupported scope. **Uncertainty/safe failure:** Report population coverage and age bands; return 'status unknown' rather than 'overdue' where the applicable regime, asset identity or evidence permission is unresolved.

ANSWER BOUNDARY

The service must not determine legal compliance, certify tests, appoint competent persons, invent due dates, authorise continued use, disclose restricted findings, or make underwriting or coverage decisions.

WHY THIS IS USEFUL

Enables the assessor to request the right competent records and target a survey where a material control lacks current, asset-specific evidence.

STAKEHOLDER QUESTION

IR-042 Which planned-maintenance backlog items have the strongest evidenced connection to material hazards, critical assets or repeated condition exceptions?

SENSORS AND TELEMETRY

Proposed evidence: CMMS backlog state linked to verified asset condition, alarm frequency, run hours, failure history and control availability, without tracking maintainer productivity or behaviour. **Coverage/timing:** All open planned-maintenance tasks on material assets and controls; daily ticket snapshots with condition summaries matched to the operating period since the task became due. **Quality gate:** where telemetry is not applicable, validate identifiers, scope, status codes, signatures, timestamp and time-zone alignment, retrieval completeness, missing records, source availability, schema drift and transformation lineage; mark calibration and sampling not applicable. Ticket descriptions may be inconsistent, some tasks are precautionary and condition signals may be unavailable or incomparable; backlog age alone does not measure consequence.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: Estates maintenance and asset owners, fire, security, water, electrical or machinery control owners, business continuity and competent maintainers. Owner-approved maintenance strategy, OEM or competent recommendations and current asset criticality lead; CMMS records show task state; telemetry supports observed condition only. **Permissions/provenance:** Retain task-to-asset-to-control lineage, restrict commercial and personnel details, and show conflicts between criticality, due date, maintenance class and current condition. **Time/staleness/conflicts:** retain action effective or version, linked observed and query retrieved times; status without a dated owner update or acceptance evidence is stale; keep open and closed records unresolved until the accountable owner confirms the state.

ANALYSIS REQUIRED

Operation type: lookup + comparison + estimate. **Interval/evidence gate:** Monthly and before a risk survey; require verified asset mapping, task scope, due date, maintenance class and hazard or criticality link for at least 95% of material backlog items. **Method:** Maintenance-backlog linkage and consequence-informed triage. Group tasks by affected control and dependency, calculate age and repeated deferral, overlay condition exceptions and test sensitivity to owner-assigned criticality rather than creating an automatic priority. **Uncertainty/safe failure:** Publish mapped share, age and evidence bands; leave items unranked when task scope or consequence is ambiguous and show how rankings change under alternative criticality classes.

ANSWER BOUNDARY

The service must not schedule work, override owner priorities, infer worker or contractor performance, diagnose condition, certify maintenance adequacy, or decide pricing, coverage or claims treatment.

WHY THIS IS USEFUL

Supports a focused maintenance challenge, specialist inspection and evidence request for backlog that may materially weaken loss controls.

IR-043

STAKEHOLDER QUESTION

Does qualified condition evidence justify competent attention to a critical asset before its scheduled inspection or maintenance date?

SENSORS AND TELEMETRY

Proposed evidence: Asset-appropriate vibration, temperature, current, pressure, flow, cycle, battery, fault and environmental signals, with local feature extraction where raw data is unnecessary. **Coverage/timing:** Owner-nominated critical assets; operating-state observations from the latest competent healthy baseline to the present, with maintenance and configuration epochs separated. **Quality gate:** confirm sensor mounting, asset identity, units and load context; retain calibration or functional checks, sample rate and aggregation, clock offset, uptime, gaps, mounting change, drift and feature lineage. Load, season, sensor movement and control changes can mimic deterioration, while many failure modes have no measurable precursor; a statistical anomaly is not a diagnosis.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: Asset owner, estates engineering, maintenance planner, OEM or competent specialist and business-continuity owner. Competent findings, OEM limits and owner-approved maintenance strategy lead; telemetry supports measured behaviour only. **Permissions/provenance:** Retain sensor siting, unit, baseline, feature and model lineage, and show disagreement between telemetry, maintenance history and specialist findings. **Time/staleness/conflicts:** retain owner-record effective, event observed and query retrieved times; live status is stale after a missed heartbeat or 15 minutes unless the owner sets a stricter limit; preserve zone and time disagreements.

ANALYSIS REQUIRED

Operation type: observation + diagnosis + recommendation. **Interval/evidence gate:** Weekly for critical assets; require a validated healthy baseline, at least 30 operating hours in comparable modes, 85% run-time coverage and no unresolved sensor-health issue. **Method:** Condition-change detection and early-attention screening. Stratify by load and mode, apply robust trend and multivariate anomaly tests, compare with post-maintenance behaviour and require persistence or independent corroboration. **Uncertainty/safe failure:** Report effect size, false-alert rate and baseline distance; use 'competent assessment indicated' and safe-fail to unknown when comparability or sensor integrity is weak.

ANSWER BOUNDARY

The service must not diagnose failure, estimate remaining life without a validated specialist method, change maintenance intervals, control equipment, certify safety or decide insurance response.

WHY THIS IS USEFUL

Gives the owner and assessor an evidence-based reason to bring forward a competent inspection without pretending that telemetry has diagnosed the asset.

IR-044

STAKEHOLDER QUESTION

Which current certificates, examination reports or formal declarations support each material control, and what asset, location, function and validity period do they actually cover?

SENSORS AND TELEMETRY

Proposed evidence: Document-register metadata, digital signatures or validation references, asset identifiers and limited telemetry linking a recorded test to the relevant device or system; document content remains permission-controlled. **Coverage/timing:** Every material control for which formal evidence is expected; evaluate at the catalogue retrieval time and retain issue, effective, expiry, test and upload dates separately. **Quality gate:** where telemetry is not applicable, validate identifiers, scope, status codes, signatures, timestamp and time-zone alignment, retrieval completeness, missing records, source availability, schema drift and transformation lineage; mark calibration and sampling not applicable. A current date does not prove the right asset or full scope, copied documents may be superseded and electronic validation may be unavailable; telemetry cannot validate professional competence.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: The issuing competent person or body, estates compliance and records owners, the relevant control owner and information governance. The original signed or independently verifiable competent record leads for its stated scope; owner registers map it to assets; telemetry may corroborate test timing only. **Permissions/provenance:** Verify through authorised references, preserve issuer, version, scope and supersession lineage, restrict detailed findings and show document-to-asset or source conflicts. **Time/staleness/conflicts:** retain permission or source effective, evidence observed and query retrieved times; apply freshness to each fact and permission, not the file as a whole; preserve competing values and the accountable owner needed to resolve them.

ANALYSIS REQUIRED

Operation type: lookup + comparison. **Interval/evidence gate:** At each risk assessment and before relying on a formal claim; require the original or verified record, identifiable issuer, stated scope, asset or zone mapping and applicable effective period. **Method:** Certificate scope, currency and asset-coverage reconciliation. Extract bounded metadata, map each document to the material control population, identify gaps, overlaps and superseded records, and sample against source-owner confirmation. **Uncertainty/safe failure:** Report percentage of controls with current, scope-matched evidence and list ambiguous mappings; safe-fail to 'formal evidence not verified' rather than assuming validity.

ANSWER BOUNDARY

The service must not certify compliance or competence, interpret legal sufficiency, alter a professional finding, expose restricted reports, declare a control safe, or decide underwriting or policy acceptance.

WHY THIS IS USEFUL

Enables the assessor to see exactly what formal evidence supports and where a new competent inspection, clarification or document is needed.

IR-045

STAKEHOLDER QUESTION

For each defect, what remedial scope and acceptance evidence were required, what was completed, and has the corrected condition remained stable?

SENSORS AND TELEMETRY

Proposed evidence: Defect, work-order and test references linked to post-work control status, fault recurrence and relevant condition telemetry, without monitoring individual workers. **Coverage/timing:** All material defects selected for closure verification; preserve issue, work, test, hand-back and follow-up periods, with observation frequency matched to the failure mode. **Quality gate:** where telemetry is not applicable, validate identifiers, scope, status codes, signatures, timestamp and time-zone alignment, retrieval completeness, missing records, source availability, schema drift and transformation lineage; mark calibration and sampling not applicable. Ticket closure does not prove the specified remedy, photographs or certificates may cover only part of the defect and stable telemetry may miss latent or intermittent failure.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: The competent defect issuer, estates maintenance, contractor manager, relevant control owner, asset custodian and records governance. The original competent defect and acceptance criteria lead; signed work and test evidence support completion; telemetry supports post-work observed condition only. **Permissions/provenance:** Retain defect-to-work-to-test-to-asset lineage, restrict commercial and personnel detail, and expose mismatched scope, dates, assets or closure states. **Time/staleness/conflicts:** retain action effective or version, linked observed and query retrieved times; status without a dated owner update or acceptance evidence is stale; keep open and closed records unresolved until the accountable owner confirms the state.

ANALYSIS REQUIRED

Operation type: lookup + observation + comparison. **Interval/evidence gate:** At proposed closure and after the owner-defined follow-up; require explicit acceptance criteria, competent evidence for each criterion and at least 30 valid days or three representative operating cycles where telemetry is relevant. **Method:** Remedial-scope verification and post-closure stability assessment. Compare specified and completed scope, verify asset identity and test result, then test recurrence and condition stability against the pre-defect and post-work baselines. **Uncertainty/safe failure:** Report criteria coverage, follow-up exposure and recurrence uncertainty; use 'closure not evidenced' or 'effectiveness not yet demonstrated' when a gate fails.

ANSWER BOUNDARY

The service must not certify remedial-work quality, close the defect, infer contractor negligence, replace competent inspection, authorise return to service, or decide claim or policy consequences.

WHY THIS IS USEFUL

Supports defensible defect close-out, targeted reinspection and challenge where paperwork and observed post-work condition do not align.

IR-046

STAKEHOLDER QUESTION

Which fire, security, water, environmental or machinery controls are isolated, bypassed, degraded or outside the owner-approved operating mode now?

SENSORS AND TELEMETRY

Proposed evidence: Command and feedback state, isolation or bypass indicators, controller mode, alarm suppression, valve position, pump availability and associated power or communications health. **Coverage/timing:** All owner-designated material controls; event capture within seconds to minutes, daily state reconciliation and explicit maintenance, test and temporary-operation windows. **Quality gate:** verify point-to-asset or zone mapping and event-code semantics; test contacts, alarms and heartbeats; retain event aggregation, clock offset, uptime, lost events, configuration drift and source-to-summary lineage. Calibrate only underlying physical sensors. Some local isolations have no remote indication, feedback can be stuck and a nonstandard mode may be legitimate; state bits do not prove field effectiveness.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: The relevant control owner, estates duty management, fire, security, water, environmental or machinery specialists, maintenance and incident coordination. Signed isolation, bypass and temporary-operation records plus competent findings lead; owner-approved normal-mode definitions follow; telemetry supports observed state only. **Permissions/provenance:** Restrict sensitive control detail, retain command-feedback, mode and authorisation lineage, and present undocumented or conflicting deviations for immediate owner resolution. **Time/staleness/conflicts:** retain owner-record effective, event observed and query retrieved times; live status is stale after a missed heartbeat or 15 minutes unless the owner sets a stricter limit; preserve zone and time disagreements.

ANALYSIS REQUIRED

Operation type: lookup + observation + comparison. **Interval/evidence gate:** Continuous for critical controls and at every shift handover; require a verified normal-mode definition, current authorisation for each deviation, named owner and healthy state evidence for at least 95% of critical controls. **Method:** Authorised-versus-observed control-state reconciliation. Compare command, feedback, mode and authorised windows, calculate undocumented duration and identify multiple dependent controls simultaneously outside normal mode. **Uncertainty/safe failure:** Report deviation duration, evidence coverage and feedback consistency; stop and escalate when a critical deviation lacks current authorisation, owner or reliable state.

ANSWER BOUNDARY

The service must not restore, isolate or command controls, decide that an abnormal mode is safe, authorise operation, certify compliance, expose restricted system details, or make insurance decisions.

WHY THIS IS USEFUL

Enables rapid clarification of the exact affected controls and supports targeted attendance before an undocumented operating deviation becomes a loss.

IR-047

STAKEHOLDER QUESTION

Is each higher-risk work activity, including hot work or intrusive isolation, covered by a live permit, mapped precautions, aftercare and accountable hand-back?

SENSORS AND TELEMETRY

Proposed evidence: Permit-state integration, temporary heat, smoke-free environmental, gas or pressure indicators where technically appropriate, fire or isolation status, access-window and aftercare check timestamps. **Coverage/timing:** The permitted work area, adjacent affected zone and declared aftercare period; event-level observations during setup, work, breaks, shutdown, fire watch or equivalent aftercare and hand-back. **Quality gate:** verify point-to-asset or zone mapping and event-code semantics; test contacts, alarms and heartbeats; retain event aggregation, clock offset, uptime, lost events, configuration drift and source-to-summary lineage. Calibrate only underlying physical sensors. Sensors cannot observe safe technique, combustible clearance, human vigilance or every ignition source; temporary devices may be moved and permit systems can lag site reality.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: Permit issuer, project or maintenance owner, contractor supervisor, fire-safety, health and safety, estates duty manager and affected local custodian. The signed permit, method statement, isolation schedule and competent site findings lead; telemetry may corroborate stated conditions and timing only. **Permissions/provenance:** Use role rather than person-level tracking, retain permit, zone, control and hand-back lineage, and expose expired, mismatched or conflicting work and aftercare states. **Time/staleness/conflicts:** retain action effective or version, linked observed and query retrieved times; status without a dated owner update or acceptance evidence is stale; keep open and closed records unresolved until the accountable owner confirms the state.

ANALYSIS REQUIRED

Operation type: lookup + comparison. **Interval/evidence gate:** Before work, continuously during the higher-risk window and at hand-back; require a current permit, mapped scope, named issuer and supervisor, stated controls and evidence covering at least 95% of the declared period. **Method:** Permit-window, control-presence and aftercare completeness assessment. Align permit validity, access, isolations, temporary controls, environmental events, breaks and aftercare checks, then calculate unpermitted or unsupported minutes. **Uncertainty/safe failure:** Report coverage and clock tolerance; safe-fail to stop-and-escalate under the existing owner procedure when permit, critical control, aftercare or current ownership is not evidenced.

ANSWER BOUNDARY

The service must not issue or approve a permit, supervise work, judge worker behaviour, certify site safety, authorise hand-back or occupation, or state that insurance acceptance is maintained.

WHY THIS IS USEFUL

Enables the assessor to and owner verify the evidence trail around a high-consequence temporary exposure and identify where competent attendance is required.

IR-048

STAKEHOLDER QUESTION

Do contractor appointment, competence, attendance, scope, isolation and completed-work records reconcile for each critical maintenance intervention?

SENSORS AND TELEMETRY

Proposed evidence: Contract and work-order references, role-level site check-in and check-out, equipment test events and completion evidence links; no continuous worker location or productivity monitoring. **Coverage/timing:** Critical maintenance tasks selected by the accountable owner; event timestamps from appointment through hand-back, limited to the work area and authorised service period. **Quality gate:** where telemetry is not applicable, validate identifiers, scope, status codes, signatures, timestamp and time-zone alignment, retrieval completeness, missing records, source availability, schema drift and transformation lineage; mark calibration and sampling not applicable. Attendance does not prove work quality, a certificate may be generic and competence records can be current yet irrelevant to the task; subcontracting may obscure responsibility.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: Procurement or contract owner, estates maintenance, permit issuer, competent service provider, local control owner and records governance. Contracts, appointments, competence records and signed service reports lead within their stated scope; work orders and telemetry corroborate timing and asset response only. **Permissions/provenance:** Minimise personal data to role and necessary identity where authorised, retain company, task, asset and evidence lineage, and show scope or provider conflicts to the contract owner. **Time/staleness/conflicts:** retain contract or source effective, external observed and local retrieved times; expired or unverifiable third-party assertions are unknown; preserve differences with local records and identify the external and internal owners.

ANALYSIS REQUIRED

Operation type: lookup + comparison. **Interval/evidence gate:** At hand-back and periodic supplier assessment; require a valid appointment, task-relevant competence evidence, asset-specific scope, attendance window and signed completion or test record. **Method:** Contractor-service evidence and asset-scope reconciliation. Link contract, work order, permit, attendance and completion by task and asset, identify missing hand-offs, unexplained time conflicts and evidence that covers a different asset or service. **Uncertainty/safe failure:** Report completeness by required evidence element and unresolved identity or scope matches; safe-fail to 'service not evidenced' rather than judging competence or quality from attendance alone.

ANSWER BOUNDARY

The service must not monitor worker performance, declare a person competent beyond the record, infer fraud or negligence, approve payment, certify work, or decide insurance or claim outcomes.

WHY THIS IS USEFUL

Supports supplier challenge, targeted reinspection and a defensible record that the right service evidence exists for the right critical asset.

IR-049

STAKEHOLDER QUESTION

Are all sensors, gateways and integrations used as risk evidence correctly mapped, commissioned, calibrated or functionally checked, time-synchronised, permissioned and healthy?

SENSORS AND TELEMETRY

Proposed evidence: Self-health, heartbeat, battery, communications quality, calibration due date, configuration version, time-sync status and test-injection results for every proposed evidence path. **Coverage/timing:** All sensors and integrations supporting a material catalogue statement; continuous health, daily quality summaries and explicit commissioning, maintenance and configuration epochs. **Quality gate:** verify service-to-asset or zone mapping and event semantics without inspecting content; retain polling and aggregation, functional tests, clock offset, uptime, log gaps, configuration drift and end-to-end provenance. Physical calibration is not applicable. A healthy heartbeat does not prove correct placement, calibration or semantic mapping, and correlated sensor failures can conceal missing evidence; manual sources may have no machine health signal.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: Telemetry service owner, estates or system asset owner, data governance, cyber security, competent calibration or commissioning provider and each authoritative source owner. Signed commissioning, calibration and mapping records lead; current owner permissions define access; health telemetry supports operational availability only, preserving effective, observed and retrieval times. **Permissions/provenance:** Retain point-to-asset-to-zone, unit, sampling, clock, transformation and permission lineage, and show any owner, mapping or configuration conflict without auto-remapping. **Time/staleness/conflicts:** retain owner-record effective, event observed and query retrieved times; live status is stale after a missed heartbeat or 15 minutes unless the owner sets a stricter limit; preserve zone and time disagreements.

ANALYSIS REQUIRED

Operation type: observation + diagnosis + comparison. **Interval/evidence gate:** Before every material answer and monthly governance assessment; require verified asset and location mapping, current permission, known units and sampling, commissioning evidence and at least 95% expected uptime for live claims. **Method:** Evidence-path qualification and health-gate assessment. Test completeness, drift, clock offset, duplicate points, stuck values, impossible ranges and end-to-end injected events, then classify each evidence path as qualified, limited or unusable. **Uncertainty/safe failure:** Publish coverage, uptime, calibration age, clock tolerance and failed checks; exclude unusable paths and return unknown rather than substituting a model or nearby sensor.

ANSWER BOUNDARY

The service must not describe proposed sensing as deployed, infer physical condition from an unqualified signal, bypass permissions, change configurations, certify calibration or system safety, or conceal evidence failures.

WHY THIS IS USEFUL

This evidence prevents apparently precise risk statements from relying on unmapped, stale, unhealthy or unauthorised telemetry.

IR-050

STAKEHOLDER QUESTION

Since the last competent test, has any material control changed state, drifted or lost observable function, and is that conclusion supported rather than inferred?

SENSORS AND TELEMETRY

Proposed evidence: Control-specific state, performance proxy, environmental context, alarm, fault, cycle and supporting power or communications health, anchored to the latest competent test window. **Coverage/timing:** Owner-selected critical controls from the test date to the next inspection or current time; event-level changes and daily summaries separated by maintenance, configuration and operating mode. **Quality gate:** for every contributing stream, verify asset and location mapping, units, calibration or functional check, sampling and aggregation, clock offset, uptime, gaps, drift and source-to-result lineage; exclude any stream that fails its gate. Telemetry observes limited proxies, a successful test covers only its stated functions and gradual deterioration may remain below detection; changed use can invalidate the baseline.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: The competent test provider, relevant control owner, estates maintenance, fire, security, water, environmental or machinery specialist and risk owner. Competent test findings and stated acceptance criteria lead; approved change and maintenance records define later context; telemetry supports observed post-test behaviour only. **Permissions/provenance:** Retain test-to-control, configuration, threshold and sensor lineage, and show conflicts between formal findings, later work and observed behaviour. **Time/staleness/conflicts:** retain owner-record effective, event observed and query retrieved times; live status is stale after a missed heartbeat or 15 minutes unless the owner sets a stricter limit; preserve zone and time disagreements.

ANALYSIS REQUIRED

Operation type: observation + comparison + diagnosis. **Interval/evidence gate:** Weekly for critical controls and before relying on an older test; require a verified test baseline, stable configuration, 30 valid days or three representative cycles and at least 90% relevant coverage. **Method:** Post-test drift, change-point and control-state surveillance. Compare matched post-test behaviour with the demonstrated test state, detect sustained shifts or new faults and require independent corroboration before escalating deterioration. **Uncertainty/safe failure:** Report effect size, baseline comparability and unobserved time; use 'test remains formal evidence, current condition uncertain' when monitoring cannot support a change statement.

ANSWER BOUNDARY

The service must not extend or cancel a certificate, certify continuing compliance, diagnose failure, alter inspection intervals, control the asset, authorise operation, or make insurance decisions.

WHY THIS IS USEFUL

Helps identify where a formally tested control needs early competent reinspection because its observed operating behaviour has materially changed.

IR-051

STAKEHOLDER QUESTION

Which spaces, assets and services are essential to continued operation, and which upstream utility, technology or supplier dependency could stop each one?

SENSORS AND TELEMETRY

Proposed evidence: Verified availability and state from power, water, cooling, ventilation, drainage, network, access, plant and environmental controls, linked to owner-approved critical assets and services. **Coverage/timing:** All services named as critical by accountable owners; one- to fifteen-minute operational state and topology snapshots after every material service or space change. **Quality gate:** verify the current asset-zone-service topology with owner sign-off; retain version and effective time, mapping completeness, geometry precision, timestamp alignment, source availability, change drift and lineage. Calibration is not applicable to topology. Telemetry may reveal technical state but not service usefulness, hidden manual dependencies and external suppliers may be absent, and criticality is an owner decision rather than a sensor output.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: Business-continuity and service owners, estates engineering, utilities, IT, security, space management, asset custodians and relevant suppliers. Current business-impact analysis and owner-approved critical-service register lead, followed by competent schematics and contracts; telemetry supports observed availability only. **Permissions/provenance:** Restrict sensitive topology and service detail, preserve service-to-dependency lineage, and show conflicts between continuity, asset and engineering records to the relevant owners. **Time/staleness/conflicts:** retain topology effective or version, supporting observed and query retrieved times; a topology becomes stale after material change; show drawing, register and telemetry differences without selecting a winner.

ANALYSIS REQUIRED

Operation type: lookup + comparison + estimate. **Interval/evidence gate:** At annual continuity assessment and after material change; require current owner-approved criticality, recovery needs and verified mapping for at least 90% of critical service dependencies. **Method:** Critical-service dependency mapping and failure-path analysis. Build a typed dependency graph, distinguish hard, capacity and manual dependencies, identify unowned edges and test single-component and selected common-cause outages. **Uncertainty/safe failure:** Report mapping coverage, unknown edges and impact ranges; safe-fail to 'dependency not established' rather than assuming independence or service continuity.

ANSWER BOUNDARY

The service must not define organisational criticality, reveal restricted infrastructure, guarantee service, control utilities or systems, replace business-impact analysis, or make underwriting decisions.

WHY THIS IS USEFUL

Helps scope resilience surveys, correct continuity assumptions and identify upstream services that need owner evidence, redundancy or contingency assessment.

IR-052

STAKEHOLDER QUESTION

For each critical load, what backup-power path and usable endurance are currently evidenced under representative demand, and which assumptions remain untested?

SENSORS AND TELEMETRY

Proposed evidence: UPS load, battery voltage, current, temperature and state estimate; generator start, output, fuel-level and alarm state; transfer-switch events and critical-circuit demand. **Coverage/timing:** Verified backup sources and mapped critical loads; one-minute trends, event-level transfers and test snapshots, with configuration and battery-replacement epochs retained. **Quality gate:** confirm meter, CT, phase and load mapping plus units and sign convention; retain calibration class or functional check, interval aggregation, clock offset, uptime, gaps, configuration drift and meter-to-summary lineage. State-of-charge and runtime estimates are model-dependent, fuel sensors can drift, actual outage load may differ from tests and shared upstream dependencies can defeat nominal redundancy.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: Estates electrical owner, UPS and generator maintainers, competent electrical specialists, critical-load owners, IT and business-continuity leads. Competent design, test and maintenance records plus owner-approved load schedules lead; OEM calculations support stated limits; telemetry supports observed load and test behaviour only. **Permissions/provenance:** Restrict critical-power topology, preserve source, load, test, model and configuration lineage, and show conflicts between design, labelled circuits and observed demand. **Time/staleness/conflicts:** retain owner-record effective, event observed and query retrieved times; live status is stale after a missed heartbeat or 15 minutes unless the owner sets a stricter limit; preserve zone and time disagreements.

ANALYSIS REQUIRED

Operation type: observation + calculation + estimate. **Interval/evidence gate:** Monthly and before continuity assessment; require current capacity and derating basis, verified load mapping, at least three representative tests or one competent full-load basis, and 95% event coverage. **Method:** Load-to-capacity reconciliation and bounded endurance estimation. Calculate load ranges, battery or fuel derating and transfer performance, simulate owner-defined demand cases and compare modelled endurance with observed tests. **Uncertainty/safe failure:** Present endurance as a range with sensitivity to load, temperature and unavailable units; return 'endurance not evidenced' when mapping, capacity or test basis is insufficient.

ANSWER BOUNDARY

The service must not guarantee runtime or start, command transfer, run tests, certify electrical safety, allocate critical loads, disclose restricted topology, or decide policy or claim response.

WHY THIS IS USEFUL

Supports targeted load testing, battery or fuel assessment, contingency planning and correction of unrealistic backup-power assumptions.

IR-053

STAKEHOLDER QUESTION

Which critical services depend on networks, gateways, cloud platforms or remote support, and what safe local function remains during each loss-of-connection mode?

SENSORS AND TELEMETRY

Proposed evidence: Network-path and gateway health, controller local-mode state, cloud or remote-service availability, buffered-data status and failover-test events; no packet payload, user content or device-content inspection. **Coverage/timing:** Owner-designated building-control and critical-service interfaces; seconds-to-minutes for availability, daily dependency summaries and explicit maintenance or configuration windows. **Quality gate:** verify service-to-asset or zone mapping and event semantics without inspecting content; retain polling and aggregation, functional tests, clock offset, uptime, log gaps, configuration drift and end-to-end provenance. Physical calibration is not applicable. Reachability does not prove functional service, a local controller may fail differently from a network outage and cyber compromise cannot be inferred from availability telemetry.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: IT network and cyber owners, estates controls, cloud or service providers, fire and security system owners, procurement and business-continuity leads. Current approved architecture, service contracts, control specifications and competent test findings lead; telemetry supports observed connectivity and mode only. **Permissions/provenance:** Do not collect content, credentials or sensitive configurations beyond need; preserve endpoint, path, mode and service-version lineage, and route architecture conflicts to accountable owners. **Time/staleness/conflicts:** retain owner-record effective, event observed and query retrieved times; live status is stale after a missed heartbeat or 15 minutes unless the owner sets a stricter limit; preserve zone and time disagreements.

ANALYSIS REQUIRED

Operation type: lookup + comparison + estimate. **Interval/evidence gate:** Quarterly and after architecture change; require a current service inventory, owner for each dependency, documented local or fallback function and at least one observed or controlled failover test for material services. **Method:** Digital-dependency and degraded-mode capability analysis. Map functions to network and remote dependencies, classify fail-open, fail-safe, local, buffered and unavailable modes, and compare declared behaviour with test or incident evidence. **Uncertainty/safe failure:** Report tested versus assumed functions and unobserved failure modes; safe-fail to 'fallback not demonstrated' where only design documentation exists.

ANSWER BOUNDARY

The service must not assess cyber vulnerability, inspect network content, expose credentials or restricted architecture, command failover, guarantee service, certify security, or decide insurance response.

WHY THIS IS USEFUL

Helps identify hidden digital single points, testable fallback assumptions and continuity actions for remotely dependent building controls.

IR-054

STAKEHOLDER QUESTION

After loss of cooling, ventilation, water, drainage or environmental control, how quickly could critical-room conditions leave the owner-approved operating envelope?

SENSORS AND TELEMETRY

Proposed evidence: Room and enclosure temperature and humidity, pressure or flow, leak and sump level, control run state, utility availability and owner-declared asset-condition requirements. **Coverage/timing:** Nominated critical rooms, assets and supporting service branches; one- to five-minute trends across normal operation, planned shutdowns or approved tests, with season and load retained. **Quality gate:** confirm representative siting, asset or zone mapping, units, traceable calibration or functional check, sampling and aggregation, clock offset, uptime, gaps, response lag, drift and raw-to-summary lineage. Observed short outages may not represent extreme weather or full load, room sensors can miss local hot spots and models depend on thermal mass, leakage and internal equipment heat assumptions.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: Asset or service owner, estates mechanical and water teams, local custodians, competent environmental or operational specialists and business-continuity owner. Owner-approved asset requirements and competent engineering studies lead; current service topology defines dependencies; telemetry supports observed condition and rate of change only. **Permissions/provenance:** Use coded sensitive spaces, preserve unit, sensor, model and service-state lineage, and show requirement, set-point or topology conflicts to the accountable owner. **Time/staleness/conflicts:** retain owner-record effective, event observed and query retrieved times; live status is stale after a missed heartbeat or 15 minutes unless the owner sets a stricter limit; preserve zone and time disagreements.

ANALYSIS REQUIRED

Operation type: observation + calculation + estimate. **Interval/evidence gate:** For continuity planning and after material change; require a current requirement, verified sensor representativeness, 14 valid operating days and either a safe controlled test or competent calibrated model. **Method:** Service-loss rate-of-change and time-to-limit estimation. Estimate condition trajectories by season and load, fit bounded rate-of-change models to outages or shutdowns, and test owner-defined service-loss durations. **Uncertainty/safe failure:** Present time-to-limit ranges with prediction intervals and assumptions; do not extrapolate beyond validated weather, load or duration, and return unknown where no safe evidence basis exists.

ANSWER BOUNDARY

The service must not infer personal health or safe occupancy, authorise a shutdown test, control HVAC or water, certify asset protection, promise response, or determine insured interruption.

WHY THIS IS USEFUL

Supports realistic response windows, sensor and contingency placement, specialist modelling and prioritisation of sensitive assets during service loss.

IR-055

STAKEHOLDER QUESTION

What recovery priorities and time objectives are owner-approved, and are they credible against current dependencies, access constraints and demonstrated restoration times?

SENSORS AND TELEMETRY

Proposed evidence: Service-state and recovery milestone telemetry, work-order and supplier response timestamps, backup-resource status and exercise observations linked to owner-defined critical services. **Coverage/timing:** All services with approved recovery objectives; incident and exercise timelines from disruption to accepted restoration, separated by scenario and operating condition. **Quality gate:** for every contributing stream, verify asset and location mapping, units, calibration or functional check, sampling and aggregation, clock offset, uptime, gaps, drift and source-to-result lineage; exclude any stream that fails its gate. Few incidents create wide uncertainty, past restoration may not represent a larger event and technical availability can precede operational acceptance.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: Business-continuity and service owners, estates, IT, security, procurement or supplier managers, incident management and senior accountable leadership. Current business-impact analysis and approved continuity plans lead; signed incident and exercise reports support demonstrated times; telemetry provides bounded operational timestamps only. **Permissions/provenance:** Restrict sensitive continuity detail, preserve objective version, scenario, dependency and milestone lineage, and show conflicts between plans, contracts and observed restoration. **Time/staleness/conflicts:** retain plan effective or version, exercise or service observed and query retrieved times; untested, superseded or out-of-date contacts and dependencies are stale; show plan-versus-observation differences and the owner needed to resolve them.

ANALYSIS REQUIRED

Operation type: lookup + comparison + estimate. **Interval/evidence gate:** At annual continuity assessment and after material incidents; require current owner-approved objectives, verified dependency paths and at least two relevant exercises or incidents, or clearly labelled scenario assumptions. **Method:** Recovery-objective plausibility and dependency-critical-path analysis. Construct critical paths, compare observed and contractual milestone distributions with objectives, model unavailable or concurrent dependencies and identify the assumptions controlling the result. **Uncertainty/safe failure:** Report observed ranges, sample size and scenario sensitivity; use 'objective not demonstrated' rather than 'unachievable' when evidence is sparse or scenarios differ.

ANSWER BOUNDARY

The service must not set or approve recovery objectives, promise recovery, allocate resources, direct incident response, value business interruption, or make underwriting, coverage or claims decisions.

WHY THIS IS USEFUL

Allows owners and assessors to challenge unsupported recovery assumptions and target exercises, contracts or resilience measures where evidence is weakest.

IR-056

STAKEHOLDER QUESTION

Which approved alternative spaces, manual procedures or temporary services can sustain essential activity, at what capacity and for how long?

SENSORS AND TELEMETRY

Proposed evidence: Availability, power, network, environmental, access and essential-equipment status for nominated alternatives, plus owner-approved booking or reservation state aggregated without personal details. **Coverage/timing:** Only alternatives named in current continuity plans; current-state snapshots, daily capacity summaries and evidence during exercises or actual activation. **Quality gate:** for every contributing stream, verify asset and location mapping, units, calibration or functional check, sampling and aggregation, clock offset, uptime, gaps, drift and source-to-result lineage; exclude any stream that fails its gate. Nominal room capacity does not prove functional suitability, shared alternatives may be unavailable during a widespread event and manual procedures can depend on unrecorded skills or supplies.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: Business-continuity and service owners, space management, estates, IT, accessibility, security, local custodians and temporary-service suppliers. Current approved continuity plans, accessibility and safety findings and owner-confirmed capacity records lead; telemetry supports observed availability and environment only. **Permissions/provenance:** Minimise booking and access data, restrict sensitive alternative locations, retain plan, space, service and exercise lineage, and expose conflicts or double-booking to the responsible owners. **Time/staleness/conflicts:** retain plan effective or version, exercise or service observed and query retrieved times; untested, superseded or out-of-date contacts and dependencies are stale; show plan-versus-observation differences and the owner needed to resolve them.

ANALYSIS REQUIRED

Operation type: lookup + calculation + estimate. **Interval/evidence gate:** At continuity assessment and before relying on an alternative; require current owner approval, mapped essential requirements, present availability, accessibility and safety evidence and a tested activation route. **Method:** Alternative-capability, capacity and activation-gap assessment. Compare required functions with alternative space, service and resource capacity, account for setup time and shared dependencies, and test selected concurrent-demand scenarios. **Uncertainty/safe failure:** Report capacity ranges, untested requirements and competing-demand assumptions; safe-fail to 'alternative not demonstrated' where a critical function or permission is missing.

ANSWER BOUNDARY

The service must not authorise relocation or occupation, determine personal suitability, reserve space, direct service changes, certify safety or accessibility, or promise continuity.

WHY THIS IS USEFUL

Supports realistic continuity planning, targeted activation exercises and owner action to close specific capacity, service or permission gaps.

IR-057

STAKEHOLDER QUESTION

What evidence shows that critical suppliers, maintainers and emergency contractors can respond within required windows during nights, weekends or widespread disruption?

SENSORS AND TELEMETRY

Proposed evidence: Call-out initiation, acknowledgement, arrival and restoration timestamps, role-level attendance, parts or resource availability status and communication-path health; no continuous worker tracking. **Coverage/timing:** Contracts and services designated critical by owners; event-level records for calls, exercises and incidents, stratified by time of day, weather, demand and access conditions. **Quality gate:** verify service-to-asset or zone mapping and event semantics without inspecting content; retain polling and aggregation, functional tests, clock offset, uptime, log gaps, configuration drift and end-to-end provenance. Physical calibration is not applicable. Contract targets are not performance guarantees, small call-out samples are unstable and arrival does not prove remedial work; subcontracting and travel disruption may be incompletely recorded.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: Procurement or contract owner, estates maintenance, service owners, suppliers, incident management and business-continuity leads. Current contracts and service-level schedules lead for commitments; signed service and incident records support observed performance; telemetry corroborates communication or asset restoration only. **Permissions/provenance:** Use company and role data where possible, preserve call-out and service lineage, restrict commercial details and show disagreements between contract, ticket and attendance records. **Time/staleness/conflicts:** retain contract or source effective, external observed and local retrieved times; expired or unverifiable third-party assertions are unknown; preserve differences with local records and identify the external and internal owners.

ANALYSIS REQUIRED

Operation type: lookup + observation + comparison. **Interval/evidence gate:** Quarterly for critical suppliers; require current contract scope, owner-required window and at least five comparable call-outs or a controlled exercise plus evidence of escalation and resource availability. **Method:** Response-time distribution, service-window and dependency assessment. Calculate acknowledgement, mobilisation, arrival and restoration intervals, stratify by adverse conditions and identify common access, parts or subcontractor dependencies. **Uncertainty/safe failure:** Report sample size, percentile ranges and censoring; do not claim capability where evidence is only contractual or where adverse scenarios remain untested.

ANSWER BOUNDARY

The service must not guarantee supplier response, rank individual workers, infer breach or negligence, approve payment, direct contractors, or decide insurance, recovery or claim consequences.

WHY THIS IS USEFUL

Enables the assessor to determine whether contracts, call-out tests, spare strategies or alternative suppliers need accountable assessment before a disruption.

IR-058

STAKEHOLDER QUESTION

Can the correct accountable roles receive, acknowledge and escalate material incident information through primary and fallback channels at all required times?

SENSORS AND TELEMETRY

Proposed evidence: Notification gateway health, role-based dispatch and acknowledgement timestamps, duty-rota validity, alternate-channel test results and local alarm or fallback status; message content and personal tracking are excluded. **Coverage/timing:** All owner-defined critical alerts and roles; event-level routing with quarterly end-to-end tests across normal, out-of-hours and degraded-communications conditions. **Quality gate:** verify point-to-asset or zone mapping and event-code semantics; test contacts, alarms and heartbeats; retain event aggregation, clock offset, uptime, lost events, configuration drift and source-to-summary lineage. Calibrate only underlying physical sensors. Delivery receipt does not prove understanding or action, personal availability changes and external networks can fail beyond observation; untested manual escalation remains an assumption.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: Incident-management, security, estates duty, fire and safety owners, IT or communications, business continuity and each service's accountable role owner. Current incident and escalation plans and approved duty arrangements lead; gateway logs support recorded dispatch and acknowledgement only, preserving plan-effective, event and retrieval times. **Permissions/provenance:** Use role addresses and minimise personal contact data, retain routing, template, channel and test lineage, restrict sensitive messages and show stale rosters or routing conflicts. **Time/staleness/conflicts:** retain owner-record effective, event observed and query retrieved times; live status is stale after a missed heartbeat or 15 minutes unless the owner sets a stricter limit; preserve zone and time disagreements.

ANALYSIS REQUIRED

Operation type: observation + comparison. **Interval/evidence gate:** Quarterly and after roster or platform change; require current role ownership, primary and fallback routes and successful test evidence for 100% of critical notification paths. **Method:** End-to-end alert-routing, acknowledgement and fallback analysis. Trace each alert from trigger through dispatch, delivery, acknowledgement and escalation, calculate timing by operating period and test single-channel failures. **Uncertainty/safe failure:** Report path success, timing ranges and untested manual steps; safe-fail to urgent owner escalation when a critical path, roster or permission cannot be verified.

ANSWER BOUNDARY

The service must not disclose private contacts or message content, infer individual responsiveness, send operational alerts, direct incident decisions, promise communication or response, or certify emergency arrangements.

WHY THIS IS USEFUL

Supports correction of stale rosters, failed gateways and untested fallback paths before they delay containment or recovery.

IR-059

STAKEHOLDER QUESTION

What have exercises and controlled tests actually demonstrated about detection, decision, fallback and recovery, and which material assumptions remain untested?

SENSORS AND TELEMETRY

Proposed evidence: Approved exercise markers, alarm and service state, role-based notification and acknowledgement, backup-resource state and recovery milestones, collected only within the agreed exercise scope. **Coverage/timing:** The systems, roles and scenarios named in the exercise plan; native event timing from exercise start through debrief, with real incidents and training injects clearly separated. **Quality gate:** for every contributing stream, verify asset and location mapping, units, calibration or functional check, sampling and aggregation, clock offset, uptime, gaps, drift and source-to-result lineage; exclude any stream that fails its gate. Participants know an exercise is occurring, simulated failures may not reproduce physical constraints and success in one scenario does not establish wider resilience.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: Exercise director, business-continuity and incident owners, estates, IT, security, fire and safety teams, service owners and independent observers. The approved exercise plan, objectives and signed debrief lead; competent test findings lead for technical functions; telemetry supports observed sequence and state only. **Permissions/provenance:** Use role identifiers, restrict sensitive scenarios and performance detail, preserve inject, event and observation lineage, and show observer or system-record conflicts. **Time/staleness/conflicts:** retain plan effective or version, exercise or service observed and query retrieved times; untested, superseded or out-of-date contacts and dependencies are stale; show plan-versus-observation differences and the owner needed to resolve them.

ANALYSIS REQUIRED

Operation type: observation + comparison. **Interval/evidence gate:** After each exercise and annually across the programme; require explicit objectives, success criteria, scoped participants and systems, preserved timeline and evidence for every claimed outcome. **Method:** Exercise-objective coverage and demonstrated-capability assessment. Map observations to objectives, distinguish tested, partially tested, assumed and failed elements, calculate detection, acknowledgement and recovery intervals, and track action closure. **Uncertainty/safe failure:** Report objective coverage and scenario limits; do not generalise beyond tested conditions and use 'not demonstrated' rather than assuming capability from plans.

ANSWER BOUNDARY

The service must not score individuals, certify resilience, direct live incident response, disclose restricted exercise weaknesses, promise performance, or make underwriting or coverage decisions.

WHY THIS IS USEFUL

Helps owners and assessors identify which resilience claims are demonstrated, which are only documented and what the next exercise should test.

IR-060

STAKEHOLDER QUESTION

Which single event, supplier, utility or shared service could disrupt several zones, buildings or critical activities at the same time?

SENSORS AND TELEMETRY

Proposed evidence: Shared-utility, network, access, supplier, environmental and central-plant status linked to owner-approved cross-zone or cross-building dependency maps. **Coverage/timing:** All declared critical activities and common services within the authorised portfolio scope; one- to fifteen-minute service state and topology snapshots after material change. **Quality gate:** verify the current asset-zone-service topology with owner sign-off; retain version and effective time, mapping completeness, geometry precision, timestamp alignment, source availability, change drift and lineage. Calibration is not applicable to topology. Portfolio mappings may use different asset definitions, external dependencies can be opaque and historical co-failure does not prove future simultaneous interruption.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: Portfolio estates and utilities, IT, security, procurement, business-continuity, building owners, local service owners and external suppliers. Current owner-approved portfolio dependency and criticality records, contracts and competent engineering studies lead; telemetry supports observed shared-service state only. **Permissions/provenance:** Restrict cross-building and security-sensitive detail, retain entity and dependency provenance, harmonise definitions explicitly and show ownership or topology conflicts. **Time/staleness/conflicts:** retain topology effective or version, supporting observed and query retrieved times; a topology becomes stale after material change; show drawing, register and telemetry differences without selecting a winner.

ANALYSIS REQUIRED

Operation type: lookup + estimate + comparison. **Interval/evidence gate:** At annual portfolio assessment and after central-service change; require a validated common-dependency map, owner-defined criticality and at least 90% mapping of in-scope critical activities. **Method:** Cross-zone common-cause and correlated-interruption scenario analysis. Construct a multi-building dependency graph, identify shared cut sets, test owner-defined outage and supplier scenarios and compare with incidents or exercises where available. **Uncertainty/safe failure:** Report affected-activity ranges, mapping gaps and scenario sensitivity; safe-fail to 'portfolio accumulation not quantifiable' where common dependencies are materially unverified.

ANSWER BOUNDARY

The service must not calculate portfolio maximum loss, decide risk appetite, underwriting, pricing or coverage, disclose restricted infrastructure, guarantee diversification, or direct continuity decisions.

WHY THIS IS USEFUL

Supports joint surveys, supplier and utility contingency assessment and clear identification of systemic dependencies that local building assessments may miss.

IR-061

STAKEHOLDER QUESTION

Which entrances, basements, roof catchments, drains and site low points are most exposed to intense rainfall, surface-water routing or local flood entry?

SENSORS AND TELEMETRY

Proposed evidence: Site rainfall, drain and gully level, sump state, water-on-floor, threshold wetness and external low-point level, linked to verified current geometry and drainage routes. **Coverage/timing:** Owner-identified entry paths, catchments and vulnerable spaces; one-minute data during rainfall, five-minute otherwise, with event inspections and official warning or rainfall records retained. **Quality gate:** confirm pipe, drain or catchment mapping, units and range; retain calibration or known-state checks, sampling and aggregation, clock offset, uptime, gaps, fouling or zero drift and alarm lineage. Point rainfall may miss local cells, sensors do not establish ground levels or hidden drainage capacity, and the May 2021 floor plan cannot prove current thresholds, routes or use.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: Estates drainage and grounds owners, building owner, competent flood or drainage surveyor, local asset custodians and the current official weather and flood-data owners. Current competent topographic, drainage and flood assessments lead; official event data leads within its location and method; telemetry supports observed local response only. **Permissions/provenance:** Retain survey version, datum, catchment, sensor and external-source provenance, respect licences, and show geometry or event-source conflicts for competent resolution. **Time/staleness/conflicts:** retain publication or model effective, observation or forecast-issued and query retrieved times; do not mix observation, forecast or model vintages silently; present source spread and send local-condition differences to the specialist owner.

ANALYSIS REQUIRED

Operation type: observation + calculation + comparison. **Interval/evidence gate:** Before seasonal risk assessment and after significant rainfall; require current surveyed entry levels and drainage routes, healthy sensors and at least 10 representative events or a competent hydraulic model. **Method:** Pluvial entry-path and rainfall-response exposure assessment. Align rainfall intensity and duration with level, wetness, pump and ingress observations, identify threshold exceedance and compare event response under known drainage condition. **Uncertainty/safe failure:** Report observed event envelope, survey tolerance and unobserved paths; do not assign a flood return period or extrapolate beyond the measured or competently modelled range.

ANSWER BOUNDARY

The service must not certify flood resistance, predict exact inundation, control pumps or barriers, establish current layout from the historical plan, determine policy response, or replace a competent flood survey.

WHY THIS IS USEFUL

Supports targeted flood survey, temporary protection, drainage intervention and evidence-led discussion of entry points most likely to need adaptation.

IR-062

STAKEHOLDER QUESTION

Which spaces, assets and services have the least verified margin during heatwaves or prolonged high internal temperature?

SENSORS AND TELEMETRY

Proposed evidence: External and zone temperature, relative humidity, solar or facade exposure proxy, cooling and ventilation state, electrical load and owner-declared asset-temperature requirements. **Coverage/timing:** Nominated heat-sensitive rooms, assets and supporting plant; one- to five-minute readings across warm-season operation, heat events and any approved controlled tests. **Quality gate:** confirm representative siting, asset or zone mapping, units, traceable calibration or functional check, sampling and aggregation, clock offset, uptime, gaps, response lag, drift and raw-to-summary lineage. Room sensors can miss local hot spots, future heat may exceed observed weather and occupancy or equipment changes can alter gains; personal heat-health risk is not assessed.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: Asset and service owners, estates mechanical and energy teams, space management, competent building-physics specialists and business-continuity owners. Owner-approved asset limits and competent thermal assessments lead; current operational records define set-points and modes; official weather data and telemetry support observed conditions only. **Permissions/provenance:** Use coded sensitive locations, retain sensor, weather, requirement and model lineage, and show conflicts between design assumptions, current use and observed temperatures. **Time/staleness/conflicts:** retain owner-record effective, event observed and query retrieved times; live status is stale after a missed heartbeat or 15 minutes unless the owner sets a stricter limit; preserve zone and time disagreements.

ANALYSIS REQUIRED

Operation type: observation + calculation + forecast. **Interval/evidence gate:** Before warm season and after material use or HVAC change; require current asset limits, representative sensor placement, at least two comparable hot events or a competently calibrated thermal model. **Method:** Heat-event thermal-margin and time-to-limit analysis. Calculate degree-hours and time above owner limits, relate internal response to external temperature, solar exposure and system state, and test owner-defined hotter-weather scenarios. **Uncertainty/safe failure:** Present margin and time-to-limit ranges with prediction intervals; safe-fail to 'future heat performance not established' when calibration, use or extreme-weather basis is weak.

ANSWER BOUNDARY

The service must not infer personal health or safe occupancy, set temperature policy, control HVAC, certify overheating compliance, guarantee asset protection, or make underwriting decisions.

WHY THIS IS USEFUL

Helps prioritise specialist modelling, cooling resilience, relocation or temporary-protection planning for the most heat-sensitive exposures.

IR-063

STAKEHOLDER QUESTION

Which external elements, openings, roof areas and exposed services have the greatest evidenced vulnerability to strong wind, wind-driven rain or debris?

SENSORS AND TELEMETRY

Proposed evidence: Local wind speed and direction where safely sited, rainfall, external door and window state, roof drainage response, facade or roof moisture indicators and external-service fault states. **Coverage/timing:** Competently selected roof, facade, opening and external-service locations; one-minute event data during storms, daily health checks and post-event inspection timestamps. **Quality gate:** verify station or model location, exposure and representativeness; retain units, quality flags, calibration or bias checks, observation or issue time, sampling and aggregation, clock basis, uptime, missing periods, drift and source-model lineage. Local wind is strongly affected by buildings, point sensors do not measure component pressure or impact, and concealed envelope defects require competent access and survey.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: Building fabric and roof owners, estates, grounds, security, local asset custodians, competent structural or facade specialists and official weather-data owners. Current competent roof, facade and structural surveys plus design records lead; official weather data leads within its measurement scope; telemetry supports local event conditions only. **Permissions/provenance:** Retain siting, exposure, unit, survey and event provenance, restrict sensitive openings and show conflicts between inspection, remedial work and observed moisture or fault evidence. **Time/staleness/conflicts:** retain owner-record effective, event observed and query retrieved times; live status is stale after a missed heartbeat or 15 minutes unless the owner sets a stricter limit; preserve zone and time disagreements.

ANALYSIS REQUIRED

Operation type: observation + comparison + estimate. **Interval/evidence gate:** Before storm season and after significant events; require current inspection evidence, verified sensor siting and at least five comparable storm events or a competent wind and envelope assessment. **Method:** Storm-event exposure, ingress-correlation and vulnerable-element screening. Align wind vector and rain with opening state, moisture, drainage and service faults, compare corrected versus uncorrected periods and identify repeated event-linked exceptions. **Uncertainty/safe failure:** Report the observed wind-rain envelope, sensor representativeness and event count; do not infer structural capacity or predict damage beyond competent design evidence.

ANSWER BOUNDARY

The service must not certify structural, roof or facade condition, authorise access, predict debris impact, assign cause or liability, control openings, or decide insurance or claims response.

WHY THIS IS USEFUL

Supports targeted roof, facade or external-service survey and evidence-based prioritisation of storm preparation and post-event inspection.

IR-064

STAKEHOLDER QUESTION

Which pipe runs, roof-level plant, entrances and drainage points could fail during freezing or snow, and what current evidence supports winter protection?

SENSORS AND TELEMETRY

Proposed evidence: External and local pipe or plant temperature, flow, heating or trace-heater state, valve and pump status, roof or entrance condition indicators and cold-weather alarm health. **Coverage/timing:** Owner-nominated exposed pipes, tanks, plant, roof areas and access points; one- to five-minute data during cold periods with pre-event checks and thaw-period monitoring. **Quality gate:** confirm pipe, drain or catchment mapping, units and range; retain calibration or known-state checks, sampling and aggregation, clock offset, uptime, gaps, fouling or zero drift and alarm lineage. Surface temperature may not represent pipe core, snow load and hidden ice require competent observation, and short historical records may not include severe cold.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: Estates mechanical, water, roof and grounds owners, security or access owner, local custodians, competent building or structural specialists and official weather-data owners. Current competent surveys, design and winterisation records lead; owner-approved operating procedures define protections; telemetry and official weather support observed conditions only. **Permissions/provenance:** Retain sensor siting, weather, protection and asset lineage, show uninstrumented exposed routes and conflicts between winter plans, work records and observed states. **Time/staleness/conflicts:** retain owner-record effective, event observed and query retrieved times; live status is stale after a missed heartbeat or 15 minutes unless the owner sets a stricter limit; preserve zone and time disagreements.

ANALYSIS REQUIRED

Operation type: observation + comparison + estimate. **Interval/evidence gate:** Before cold forecasts and after freeze events; require current exposed-asset mapping, protection requirements, healthy local sensors and at least 10 cold nights or a competent thermal basis. **Method:** Cold-exposure duration, protection-state and freeze-risk screening. Calculate hours below owner-defined thresholds, concurrent flow and heat-trace state, thaw cycles and access exceptions, then test owner-defined colder-duration scenarios. **Uncertainty/safe failure:** Report threshold uncertainty, unobserved routes and scenario range; safe-fail to unknown where concealed condition, snow load or protection performance requires physical inspection.

ANSWER BOUNDARY

The service must not certify frost protection, structural snow capacity or safe access, control heating or valves, predict burst timing, infer personal risk, or decide insurance response.

WHY THIS IS USEFUL

Helps target winterisation checks, pipe and roof survey, temporary protection and continuity planning for exposed services.

IR-065

STAKEHOLDER QUESTION

How would drought, water restrictions or reduced supply resilience affect cooling, sanitation, fire protection and other critical services, and where is the dependency untested?

SENSORS AND TELEMETRY

Proposed evidence: Incoming pressure and flow, storage or tank level, cooling and operational-water demand, fire-water availability status, leak losses and owner-approved demand curtailment indicators. **Coverage/timing:** Verified water supplies, storage, critical branches and nominated services; one- to fifteen-minute operational data, daily demand summaries and separate restriction or low-pressure periods. **Quality gate:** confirm pipe, drain or catchment mapping, units and range; retain calibration or known-state checks, sampling and aggregation, clock offset, uptime, gaps, fouling or zero drift and alarm lineage. Metering may not separate all uses, future restrictions are externally determined and fire-water adequacy cannot be established from ordinary demand data alone.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: Estates water and mechanical owners, fire-safety, local service owners, business continuity, utility liaison and the current external water-supply data owner. Current supply agreements, competent fire-water and engineering findings and approved continuity plans lead; utility notices lead for declared restrictions; telemetry supports observed demand and pressure only. **Permissions/provenance:** Retain supply, branch, unit and notice provenance, restrict critical-service details and show conflicts between design, contracts, owner priorities and observed use. **Time/staleness/conflicts:** retain owner-record effective, event observed and query retrieved times; live status is stale after a missed heartbeat or 15 minutes unless the owner sets a stricter limit; preserve zone and time disagreements.

ANALYSIS REQUIRED

Operation type: lookup + estimate + comparison. **Interval/evidence gate:** At annual resilience assessment and during restrictions; require verified topology, current critical demand, usable storage basis and 12 representative months or a competent demand model. **Method:** Demand-supply stress, storage endurance and service-priority scenario analysis. Construct demand bands by operating mode, estimate storage or reduced-supply endurance, test leak and curtailment scenarios and identify services sharing constrained branches. **Uncertainty/safe failure:** Present endurance and unmet-demand ranges with sensitivity to demand and supply assumptions; return 'fire protection requires competent confirmation' where specialist evidence is needed.

ANSWER BOUNDARY

The service must not certify fire-water adequacy, allocate scarce water, control valves or services, promise supply, infer health effects, or make policy, underwriting or claims decisions.

WHY THIS IS USEFUL

Supports continuity planning, leak reduction, storage or supply assessment and early discussion of which critical services need competent drought contingencies.

IR-066

STAKEHOLDER QUESTION

Across the remaining life of major assets, how do credible climate scenarios change flood, heat, wind, cold, moisture and water-supply exposure?

SENSORS AND TELEMETRY

Proposed evidence: Commissioned site weather and condition telemetry used to characterise the present baseline, not to generate future climate; current asset age, condition and performance indicators provide local context. **Coverage/timing:** Owner-selected assets and hazards over explicit near-, mid- and long-term horizons; baseline periods must state coverage and be aligned with appropriately scaled official climate projections. **Quality gate:** verify station or model location, exposure and representativeness; retain units, quality flags, calibration or bias checks, observation or issue time, sampling and aggregation, clock basis, uptime, missing periods, drift and source-model lineage. Site records are short, climate projections are scenario-dependent and downscaling does not resolve building-level construction or operational vulnerability without competent modelling.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: Building and asset owners, estates strategy, sustainability, business continuity, competent climate, flood, structural and building-physics specialists, and official projection-data owners. Current official climate projections and competent hazard studies lead for future scenarios; owner asset plans define life and change assumptions; telemetry supports the present observed baseline only. **Permissions/provenance:** Retain scenario, model, vintage, spatial scale, asset-life and transformation lineage, respect external licences and show conflicts between projection sources and owner assumptions. **Time/staleness/conflicts:** retain publication or model effective, observation or forecast-issued and query retrieved times; do not mix observation, forecast or model vintages silently; present source spread and send local-condition differences to the specialist owner.

ANALYSIS REQUIRED

Operation type: forecast + calculation + comparison. **Interval/evidence gate:** At long-term asset assessment; require at least two credible scenarios, explicit time horizons, current asset life and condition, competent hazard translation and a documented present baseline. **Method:** Scenario-to-asset exposure overlay and time-horizon sensitivity analysis. Translate scenario variables into hazard metrics through competent methods, compare them with design or operating envelopes and test adaptation and no-adaptation cases. **Uncertainty/safe failure:** Present scenario ranges, model spread and unsupported local detail; do not collapse projections into a single forecast or extrapolate beyond the specialist method's scope.

ANSWER BOUNDARY

The service must not predict future weather or loss, certify climate resilience, choose a scenario as fact, set asset strategy, calculate insured loss, or make underwriting, pricing or coverage decisions.

WHY THIS IS USEFUL

Helps owners and assessors identify which long-lived assets need specialist climate assessment and when adaptation evidence should enter capital planning.

IR-067

STAKEHOLDER QUESTION

Could current energy-price, supply or demand pressure lead to operating changes that weaken temperature control, maintenance, lighting, security or other loss controls?

SENSORS AND TELEMETRY

Proposed evidence: Aggregate energy demand, control schedules and set-point changes, HVAC and lighting run state, backup-power use, maintenance deferrals and owner-declared curtailment or demand-response events. **Coverage/timing:** Material loss-control services and affected zones; 15-minute energy and state data with explicit pre-change, change and post-change periods, separated by season and occupancy mode. **Quality gate:** confirm meter, CT, phase and load mapping plus units and sign convention; retain calibration class or functional check, interval aggregation, clock offset, uptime, gaps, configuration drift and meter-to-summary lineage. Energy reduction can be benign, similar load changes can arise from weather or use, and telemetry cannot reveal informal budget decisions or maintenance quality.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: Energy and estates operations, finance or budget owners, security, fire and safety control owners, maintenance, local service owners and business continuity. Approved operating policy, change-control records, budgets and competent control requirements lead; telemetry supports observed demand, schedules and conditions only. **Permissions/provenance:** Retain change, set-point, tariff or event and control lineage, restrict commercially sensitive cost detail and show unexplained operational changes to accountable owners. **Time/staleness/conflicts:** retain owner-record effective, event observed and query retrieved times; live status is stale after a missed heartbeat or 15 minutes unless the owner sets a stricter limit; preserve zone and time disagreements.

ANALYSIS REQUIRED

Operation type: lookup + estimate + comparison. **Interval/evidence gate:** Monthly and after curtailment or major tariff change; require an approved change record, 28 comparable pre- and post-change days and at least 90% state and environmental coverage. **Method:** Energy-operation change reconciliation and loss-control impact screening. Normalise for weather and operating mode, compare control availability, environmental excursions and maintenance state, and test whether changes coincide with owner-defined protection limits. **Uncertainty/safe failure:** Report adjusted effect ranges and confounding factors; return 'impact not attributable' when weather, use or change records are insufficient.

ANSWER BOUNDARY

The service must not infer financial distress, set energy policy, identify individual behaviour, certify controls, reverse operating changes, or make underwriting, pricing, coverage or claims decisions.

WHY THIS IS USEFUL

Supports accountable assessment where efficiency or cost actions may unintentionally reduce loss-control performance or resilience.

IR-068

STAKEHOLDER QUESTION

Have decarbonisation, insulation, ventilation, heating or controls projects introduced new fire, moisture, overheating, electrical, interface or maintenance exposure?

SENSORS AND TELEMETRY

Proposed evidence: Pre- and post-project temperature, humidity, moisture, ventilation and electrical-load patterns, control modes, alarm interfaces and fault or maintenance events, mapped to the changed fabric and systems. **Coverage/timing:** The project scope, adjacent affected zones and supporting services; commissioning-period data plus at least one representative season before and after where available. **Quality gate:** confirm representative siting, asset or zone mapping, units, traceable calibration or functional check, sampling and aggregation, clock offset, uptime, gaps, response lag, drift and raw-to-summary lineage. Building use and weather may change between periods, sensors cannot identify concealed workmanship and a successful commissioning test may not cover seasonal or long-term effects.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: Project sponsor, designers and contractors, estates, fire and safety owners, sustainability, building fabric and services owners, and competent commissioning specialists. Approved design, change-control, fire and safety assessments and competent commissioning findings lead; telemetry supports observed pre- and post-change conditions only. **Permissions/provenance:** Retain design version, affected asset, commissioning, baseline and model lineage, restrict proprietary design details and show conflicts between project intent, hand-back and current operation. **Time/staleness/conflicts:** retain owner-record effective, event observed and query retrieved times; live status is stale after a missed heartbeat or 15 minutes unless the owner sets a stricter limit; preserve zone and time disagreements.

ANALYSIS REQUIRED

Operation type: lookup + observation + comparison. **Interval/evidence gate:** At hand-back and seasonal assessment; require current as-built scope, competent commissioning evidence, verified sensor mapping and at least 60 valid comparable days or a specialist model for unavailable seasons. **Method:** Project-change hazard assessment and controlled pre-post performance comparison. Map each physical and control change to plausible hazards, compare matched weather and use periods, inspect sustained moisture, heat, electrical and fault shifts, and refer material findings to specialists. **Uncertainty/safe failure:** Report effect ranges and baseline comparability; use 'new exposure not ruled out' where concealed construction, season or operating change is unobserved.

ANSWER BOUNDARY

The service must not certify project compliance or safety, infer workmanship defects, approve hand-back, control systems, attribute damage cause, or decide insurance or claims consequences.

WHY THIS IS USEFUL

Helps target post-project survey, seasonal commissioning, defect investigation and risk-register updates around unintended consequences of transition work.

IR-069

STAKEHOLDER QUESTION

Where photovoltaic generation, battery storage or electric-vehicle charging is present, what current fire, electrical, concentration and continuity evidence is needed?

SENSORS AND TELEMETRY

Proposed evidence: Inverter, charger and battery voltage, current, temperature, state, fault and isolation indicators; local ventilation or environment, protection-device status and approved fire-interface states. **Coverage/timing:** All owner-declared generation, storage, charging equipment, associated rooms, external areas and critical loads; seconds-to-five-minute operation with test, firmware and configuration epochs retained. **Quality gate:** confirm meter, CT, phase and load mapping plus units and sign convention; retain calibration class or functional check, interval aggregation, clock offset, uptime, gaps, configuration drift and meter-to-summary lineage. Manufacturer state estimates are model-dependent, remote status does not reveal physical damage or installation quality and battery thermal events may develop locally beyond sensor coverage.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: Estates electrical and energy owners, fire-safety, competent installers and maintainers, network or supply-interface owner, security, business continuity and local asset custodians. Approved design, fire strategy, competent installation and test records and OEM requirements lead; telemetry supports observed operating condition only. **Permissions/provenance:** Restrict detailed critical-power and security locations, retain device, firmware, protection, test and asset lineage, and show conflicts between registers, labels and observed configuration. **Time/staleness/conflicts:** retain owner-record effective, event observed and query retrieved times; live status is stale after a missed heartbeat or 15 minutes unless the owner sets a stricter limit; preserve zone and time disagreements.

ANALYSIS REQUIRED

Operation type: lookup + observation + comparison. **Interval/evidence gate:** Monthly and after fault, firmware or configuration change; require current as-built mapping, competent test evidence, OEM limits and at least 95% event coverage for critical protection and temperature states. **Method:** Energy-system operating-envelope, protection and concentration assessment. Compare operation with approved limits, identify repeated faults, simultaneous charging or storage concentration, shared dependencies and demonstrated isolation or fallback during tests. **Uncertainty/safe failure:** Report unobserved devices, measurement tolerance and model dependence; stop and escalate owner-rule breaches, otherwise avoid diagnosing battery or electrical condition.

ANSWER BOUNDARY

The service must not certify electrical or fire safety, command charging, isolation or storage, infer battery health beyond evidence, expose restricted topology, or decide underwriting, pricing, coverage or claims.

WHY THIS IS USEFUL

Supports specialist electrical and fire survey, protection testing, concentration assessment and continuity planning for newer energy assets.

IR-070

STAKEHOLDER QUESTION

Which adaptation options merit competent appraisal first, what measurable trigger would justify action, and how would reduced exposure be verified?

SENSORS AND TELEMETRY

Proposed evidence: Relevant hazard and control telemetry, such as flood level, heat margin, moisture, drainage response, backup capacity and service availability, used only to measure present conditions and post-action change. **Coverage/timing:** Owner-selected hazards, assets and time horizons; baseline and post-action monitoring periods defined in advance, with external scenario and asset-life data linked through provenance. **Quality gate:** for every contributing stream, verify asset and location mapping, units, calibration or functional check, sampling and aggregation, clock offset, uptime, gaps, drift and source-to-result lineage; exclude any stream that fails its gate. Short monitoring cannot prove long-term benefit, rare events may not recur during evaluation and one intervention can shift exposure elsewhere; costs and organisational priorities are owner inputs.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: Building and asset owners, estates strategy, finance, sustainability, business continuity, competent hazard specialists, insurers or risk engineers as advisers and official scenario-data owners. Owner risk criteria, competent studies, approved asset plans and official scenarios lead; telemetry supports baseline and observed post-action performance only. **Permissions/provenance:** Retain option, trigger, scenario, cost-source, assumption and measurement-plan lineage, restrict commercially sensitive plans and show disagreements rather than forcing a single priority. **Time/staleness/conflicts:** retain owner-record effective, event observed and query retrieved times; live status is stale after a missed heartbeat or 15 minutes unless the owner sets a stricter limit; preserve zone and time disagreements.

ANALYSIS REQUIRED

Operation type: comparison + estimate + recommendation. **Interval/evidence gate:** During asset and risk planning; require at least three feasible options including no action, explicit owner criteria, two climate or hazard scenarios, implementation constraints and a measurable verification plan. **Method:** Robust-options, trigger-threshold and monitored-benefit analysis. Compare options using multi-criteria and regret analysis, test sensitivity to hazard, cost and asset-life assumptions, and define observable trigger and post-action indicators before recommendation discussion. **Uncertainty/safe failure:** Present dominance, trade-offs and scenario ranges rather than one optimal answer; safe-fail to 'specialist option appraisal required' where benefits or adverse effects cannot be evidenced.

ANSWER BOUNDARY

The service must not approve capital work, set risk appetite, promise loss reduction, certify adaptation, calculate a premium benefit, or make underwriting, pricing, coverage or claims decisions.

WHY THIS IS USEFUL

Gives owners and assessors a transparent basis for specialist option appraisal, monitoring design and timely escalation when exposure crosses an agreed trigger.

IR-071

STAKEHOLDER QUESTION

For this answer, what share of relevant assets, locations and time periods is covered by healthy, correctly mapped evidence, and what remains unseen?

SENSORS AND TELEMETRY

Proposed evidence: Source inventory, asset and zone mapping, heartbeat, uptime, battery, calibration, sampling, time-sync and integration-health metadata for every evidence path used. **Coverage/timing:** The exact asset, location and decision-period population stated in the question; evaluate at retrieval time and retain historical health for the claimed observation window. **Quality gate:** where telemetry is not applicable, validate identifiers, scope, status codes, signatures, timestamp and time-zone alignment, retrieval completeness, missing records, source availability, schema drift and transformation lineage; mark calibration and sampling not applicable. Health does not prove correct siting or semantics, multiple points may share one failure and manual or restricted evidence may lack automated health measures.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: Telemetry service owner, each physical system and asset owner, data governance, competent commissioning or calibration providers and the requesting assessor. Signed mapping, commissioning and calibration records lead; current owner registers define the population; health telemetry supports availability only. **Permissions/provenance:** Retain point-to-asset-to-zone, unit, sampling, clock and permission lineage, identify shared failure domains and show unmapped or conflicting sources explicitly. **Time/staleness/conflicts:** retain permission or source effective, evidence observed and query retrieved times; apply freshness to each fact and permission, not the file as a whole; preserve competing values and the accountable owner needed to resolve them.

ANALYSIS REQUIRED

Operation type: calculation + comparison. **Interval/evidence gate:** Before every material answer; require a declared denominator, verified mapping, purpose-specific freshness rule and health evidence for each included source. **Method:** Asset-location-time coverage and evidence-path qualification analysis. Calculate asset, spatial and temporal coverage separately, weight critical gaps without hiding the raw percentages, and remove duplicated or non-independent sources from corroboration counts. **Uncertainty/safe failure:** Publish coverage percentages, longest gap, evidence age and shared-failure share; safe-fail to unknown when the denominator or mapping is materially uncertain.

ANSWER BOUNDARY

The service must not equate high coverage with truth or control adequacy, treat unobserved areas as normal, bypass restricted sources, invent denominator values, or make insurance decisions.

WHY THIS IS USEFUL

Enables the assessor to judge whether an answer is sufficiently observed, needs a physical survey or must be limited to a smaller scope.

IR-072

STAKEHOLDER QUESTION

When schedules, registers, certificates, tickets, inspections and telemetry conflict, which source is authoritative for each fact and how does disagreement limit the answer?

SENSORS AND TELEMETRY

Proposed evidence: Source-version, effective-date, observation-time, retrieval-time, ownership and lineage metadata, plus conflict flags generated without changing the underlying records. **Coverage/timing:** Every material fact with two or more candidate sources; assess at the decision time and retain the historical version used in any earlier answer. **Quality gate:** where telemetry is not applicable, validate identifiers, scope, status codes, signatures, timestamp and time-zone alignment, retrieval completeness, missing records, source availability, schema drift and transformation lineage; mark calibration and sampling not applicable. A newer source is not automatically more authoritative, sources can govern different scopes and a conflict may be semantic rather than factual.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: Policy or claims custodian, competent professional, accountable asset or control owner, source-system custodian, data governance and the requesting assessor. Policy terms lead for contractual scope, competent findings for stated technical findings, current accountable-owner records for declared arrangements and qualified telemetry for observed conditions only. **Permissions/provenance:** Record source owner, authority basis, scope, version, all three times and transformation lineage; keep conflicting values visible and route unresolved authority questions to accountable owners. **Time/staleness/conflicts:** retain permission or source effective, evidence observed and query retrieved times; apply freshness to each fact and permission, not the file as a whole; preserve competing values and the accountable owner needed to resolve them.

ANALYSIS REQUIRED

Operation type: lookup + comparison. **Interval/evidence gate:** For every conflict affecting a material statement; require identified owners, comparable definitions, exact scope and timestamps before applying precedence. **Method:** Fact-level precedence, semantic alignment and conflict-impact analysis. Classify conflicts as scope, time, identity, unit, status or authority differences, apply the documented precedence rule and test which conclusions change under each unresolved value. **Uncertainty/safe failure:** Report resolved and unresolved conflicts and conclusion sensitivity; safe-fail to conflicting where no source has clear authority for the exact fact and period.

ANSWER BOUNDARY

The service must not silently merge or overwrite sources, treat telemetry as authoritative by recency, interpret policy wording, alter professional findings, or decide coverage, liability or claims.

WHY THIS IS USEFUL

This evidence produces a transparent evidence trail and directs the disagreement to the owner best placed to resolve it before reliance.

IR-073

STAKEHOLDER QUESTION

Are clocks, time zones and event timestamps aligned well enough to support the claimed incident sequence, duration, response time or control overlap?

SENSORS AND TELEMETRY

Proposed evidence: Clock-source, offset, drift, synchronisation, time-zone, daylight-saving and ingestion-latency metadata, together with shared physical anchor events visible in more than one source. **Coverage/timing:** Every source used in the claimed sequence and the full event window; preserve native timestamps and clock-health history rather than rewriting source records. **Quality gate:** where telemetry is not applicable, validate identifiers, scope, status codes, signatures, timestamp and time-zone alignment, retrieval completeness, missing records, source availability, schema drift and transformation lineage; mark calibration and sampling not applicable. Some systems record send rather than occurrence time, manual times may be coarsened and network delay can vary; a shared label does not prove a shared event.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: Each source-system custodian, IT time-service owner, evidence custodian, incident owner and appointed competent investigator. Native source records and documented clock configurations lead for recorded time; competent investigation determines evidential interpretation. **Permissions/provenance:** Retain native and normalised times, offset method, anchor-event lineage and uncertainty, and show clocks or time zones that cannot be reconciled. **Time/staleness/conflicts:** retain permission or source effective, evidence observed and query retrieved times; apply freshness to each fact and permission, not the file as a whole; preserve competing values and the accountable owner needed to resolve them.

ANALYSIS REQUIRED

Operation type: calculation + comparison. **Interval/evidence gate:** Before any sub-minute or response-time claim; require known time zone, at least two credible shared anchors or an independently logged offset, and clock uncertainty smaller than the claimed ordering gap. **Method:** Clock-offset, drift and interval-ordering analysis. Estimate offset and drift with robust regression, propagate timestamp intervals through the sequence and test whether alternative clock corrections reverse event order. **Uncertainty/safe failure:** Report each event as a time or interval with tolerance; safe-fail to 'order unresolved' whenever intervals overlap or clock evidence is inadequate.

ANSWER BOUNDARY

The service must not edit source timestamps, manufacture precision, infer unrecorded actions, declare response compliance, assign fault, or make claims or coverage decisions from uncertain timing.

WHY THIS IS USEFUL

This evidence prevents false causal or performance conclusions created by apparently precise but misaligned system clocks.

IR-074

STAKEHOLDER QUESTION

What baseline and threshold define normal for this asset or zone, and are they valid for the present season, use, load and control configuration?

SENSORS AND TELEMETRY

Proposed evidence: The relevant qualified condition or state signals, plus operating mode, load, weather, maintenance and configuration markers needed to form comparable baseline groups. **Coverage/timing:** The exact asset or zone and operating states to which the threshold will apply; retain baseline, validation and current periods separately at native or justified aggregate resolution. **Quality gate:** for every contributing stream, verify asset and location mapping, units, calibration or functional check, sampling and aggregation, clock offset, uptime, gaps, drift and source-to-result lineage; exclude any stream that fails its gate. Historical data can include faults, seasonal use changes and sensor drift, while owner set-points are not always risk thresholds; rare modes may lack enough examples.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: Asset or control owner, competent specialist, estates operations, local service owner, telemetry owner and data governance. Competent limits and owner-approved operating envelopes lead; validated historical data supports empirical baselines only. **Permissions/provenance:** Retain baseline inclusion rules, threshold version, units, sensor and configuration lineage, and show conflicts between formal limits, set-points and empirical behaviour. **Time/staleness/conflicts:** retain owner-record effective, event observed and query retrieved times; live status is stale after a missed heartbeat or 15 minutes unless the owner sets a stricter limit; preserve zone and time disagreements.

ANALYSIS REQUIRED

Operation type: observation + comparison + estimate. **Interval/evidence gate:** Before anomaly use and after material change; require at least 30 valid days or 20 representative cycles per state, 85% coverage, stable sensor health and independent validation data. **Method:** Context-stratified baseline validation and threshold calibration. Exclude known faults, stratify by season, load and mode, use robust quantiles or specialist limits, and validate sensitivity, false-alert rate and drift on held-out periods. **Uncertainty/safe failure:** Publish baseline range, sample size, false-alert estimate and excluded states; safe-fail to 'no valid baseline' rather than borrowing an unrelated asset or season.

ANSWER BOUNDARY

The service must not set operational or safety thresholds, change controls, certify normal condition, extrapolate to unobserved modes, or translate an anomaly threshold into an insurance decision.

WHY THIS IS USEFUL

This evidence makes anomaly statements challengeable and prevents normal seasonal or operational change from being presented as deterioration.

STAKEHOLDER QUESTION

IR-075

Which assumptions dominate this loss, resilience or climate scenario, and how far does the bounded conclusion move across plausible alternatives?

SENSORS AND TELEMETRY

Proposed evidence: Qualified observed data used to estimate present-state inputs, with explicit ranges for unobserved rates, capacities, durations and dependencies; no model output becomes a fact source. **Coverage/timing:** The owner-defined scenario, assets, hazards, time horizon and decision interval; every input retains its source period and transformation from evidence to model parameter. **Quality gate:** verify the current asset-zone-service topology with owner sign-off; retain version and effective time, mapping completeness, geometry precision, timestamp alignment, source availability, change drift and lineage. Calibration is not applicable to topology. Input ranges may reflect judgement rather than frequency, dependencies can invalidate simple sampling and model structure may dominate numerical uncertainty.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: Scenario sponsor, accountable asset and risk owners, competent hazard specialists, data owners, finance or continuity advisers and the requesting assessor. Competent studies and current owner records lead for inputs within scope; observed telemetry supports present conditions; assumptions remain labelled. **Permissions/provenance:** Retain model version, equations or rules, parameter sources, dependence assumptions and permissions; show conflicting expert ranges and do not average away distinct views. **Time/staleness/conflicts:** retain permission or source effective, evidence observed and query retrieved times; apply freshness to each fact and permission, not the file as a whole; preserve competing values and the accountable owner needed to resolve them.

ANALYSIS REQUIRED

Operation type: calculation + estimate + comparison. **Interval/evidence gate:** Before communicating a consequential scenario; require an explicit model, source or rationale for every material input, plausible ranges, dependence assumptions and at least two alternative cases. **Method:** Local and global sensitivity, uncertainty propagation and scenario comparison. Apply one-at-a-time screening, then Monte Carlo or structured scenario sampling where justified, rank inputs by contribution to output variance and test model-form alternatives. **Uncertainty/safe failure:** Present distributions or ranges, sensitivity indices and structural caveats; do not assign probabilities to judgement-only scenarios or report false precision.

ANSWER BOUNDARY

The service must not choose policy terms, risk appetite or underwriting action, present a model as prediction, hide assumptions, optimise premiums, or decide pricing, coverage, claim value or liability.

WHY THIS IS USEFUL

This evidence shows stakeholders which evidence or specialist judgement would most reduce uncertainty and which conclusions remain robust despite it.

STAKEHOLDER QUESTION

IR-076

How should a rare but severe hazard be stated when local incidents, telemetry and exposure history are sparse or much shorter than the relevant return period?

SENSORS AND TELEMETRY

Proposed evidence: Qualified local condition and event observations used to establish what has actually been seen, plus sensor detection limits and unobserved time; absence is recorded, not interpreted as absence of hazard. **Coverage/timing:** The defined hazard, site and observation period, supplemented by appropriately scoped external records and competent studies whose spatial and temporal relevance is explicit. **Quality gate:** verify station or model location, exposure and representativeness; retain units, quality flags, calibration or bias checks, observation or issue time, sampling and aggregation, clock basis, uptime, missing periods, drift and source-model lineage. Short records cannot estimate extreme frequency, external loss data may not match construction or exposure and no local event may simply reflect limited observation.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: Competent hazard specialists, building and asset owners, incident and claims-data custodians, official hazard-data owners and the requesting assessor. Competent site assessments and official hazard evidence lead for severe-hazard characterisation; local telemetry supports only observed conditions and detection limits. **Permissions/provenance:** Respect restricted loss-data licences, retain source population, period and transfer assumptions, and show incompatible datasets or expert views rather than pooling them blindly. **Time/staleness/conflicts:** retain permission or source effective, evidence observed and query retrieved times; apply freshness to each fact and permission, not the file as a whole; preserve competing values and the accountable owner needed to resolve them.

ANALYSIS REQUIRED

Operation type: forecast + estimate + comparison. **Interval/evidence gate:** For every low-frequency high-consequence statement; require a physical mechanism, competent hazard basis, explicit observation exposure and at least one independent external evidence source. **Method:** Evidence-bounded rare-event scenario and upper-bound analysis. Separate plausibility, observed frequency and consequence, use zero-event bounds only with a valid exposure denominator, and test severe but credible scenarios without assigning unjustified probabilities. **Uncertainty/safe failure:** Report observation length, detection limit and scenario range; never convert no recorded event into zero risk or a precise return period.

ANSWER BOUNDARY

The service must not predict a rare event, certify resilience, invent frequency, disclose restricted claims data, calculate probable maximum loss, or decide underwriting, pricing or coverage.

WHY THIS IS USEFUL

Allows severe hazards to remain visible and honestly qualified while directing the assessor toward specialist evidence rather than false reassurance.

IR-077

STAKEHOLDER QUESTION

Can a competent assessor reproduce and challenge every material statement from its sources, methods, assumptions, permissions, coverage and limitations?

SENSORS AND TELEMETRY

Proposed evidence: Automated provenance, source-version, mapping, health, transformation and access records for each evidence item and derived output, without altering the authoritative source. **Coverage/timing:** Every material statement in the catalogue answer; capture provenance at query time and preserve the exact source versions and method configuration used. **Quality gate:** where telemetry is not applicable, validate identifiers, scope, status codes, signatures, timestamp and time-zone alignment, retrieval completeness, missing records, source availability, schema drift and transformation lineage; mark calibration and sampling not applicable. A complete lineage record does not prove source truth, proprietary tools may limit reproducibility and restricted inputs may allow only controlled verification.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: Each source owner, telemetry and analytics owners, data governance, information security, competent assessor and records custodian. Authoritative sources retain their own status; provenance describes how they were used, not their correctness throughout. **Permissions/provenance:** Record question, purpose, owner, source IDs, versions, mappings, units, filters, joins, models, assumptions, permissions and output hash, with controlled routes for restricted verification. **Time/staleness/conflicts:** retain permission or source effective, evidence observed and query retrieved times; apply freshness to each fact and permission, not the file as a whole; preserve competing values and the accountable owner needed to resolve them.

ANALYSIS REQUIRED

Operation type: lookup + calculation + comparison. **Interval/evidence gate:** For every material answer; require complete lineage for all supporting statements, a versioned method and an independent rerun or spot check before external reliance. **Method:** Claim-evidence graph construction and reproducibility testing. Build a directed claim-evidence graph, rerun derived values from preserved inputs, compare hashes or tolerances and identify orphan claims, unused evidence and undocumented transformations. **Uncertainty/safe failure:** Report lineage completeness and reproducibility status; safe-fail to 'not reproducible' when a material input, method step or permission record is missing.

ANSWER BOUNDARY

The service must not claim absolute truth, expose restricted inputs, modify authoritative records, conceal proprietary limitations, certify professional findings, or make consequential insurance decisions.

WHY THIS IS USEFUL

Gives assessors an inspectable evidence trail and makes correction possible without relying on an opaque dashboard or unrepeatable analysis.

IR-078

STAKEHOLDER QUESTION

How can necessary risk evidence be shared among owners, insurers, engineers and claims specialists without disclosing personal, security-sensitive or restricted information?

SENSORS AND TELEMETRY

Proposed evidence: Data-classification, permission, audience, aggregation, redaction and access-control metadata; physical telemetry is queried only after the minimum fields and time window are approved. **Coverage/timing:** Each proposed audience and decision purpose; views are generated at request time because role, investigation status, sensitivity and source permissions may change. **Quality gate:** where telemetry is not applicable, validate identifiers, scope, status codes, signatures, timestamp and time-zone alignment, retrieval completeness, missing records, source availability, schema drift and transformation lineage; mark calibration and sampling not applicable. Aggregation can still reveal small groups or sensitive locations, redaction can remove necessary context and prior sharing does not authorise reuse for another purpose.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: Information governance, security, legal or records owner, source custodians, policy or claims custodian, accountable building owner and the receiving organisation's sponsor. Law, policy terms, contracts, investigation controls and current source-owner permissions lead; no analytic convenience overrides restriction. **Permissions/provenance:** Apply audience-specific views, minimum necessary fields, coded locations, small-number suppression, secure delivery, retention limits and complete access and transformation lineage. **Time/staleness/conflicts:** retain permission effective and expiry, event observed and query retrieved times; operational status follows the owner's short freshness limit; preserve differences in minimised form without exposing restricted detail.

ANALYSIS REQUIRED

Operation type: lookup + comparison + recommendation. **Interval/evidence gate:** Before every external or cross-role share; require a stated purpose, named role, authority basis, necessary claims, source-owner approval, secure route and expiry or retention rule. **Method:** Disclosure-risk, utility and tiered-view design assessment. Map each recipient need to minimum evidence, test personal and security disclosure risk, generate restricted and summary tiers and verify the delivered view against the approval record. **Uncertainty/safe failure:** State what is withheld and how that limits interpretation without revealing content; safe-fail to no share where purpose, authority or secure route is unresolved.

ANSWER BOUNDARY

The service must not expose identities, access patterns, security layouts, credentials, privileged material or commercially restricted records, infer withheld content, or decide insurance entitlement.

WHY THIS IS USEFUL

Enables useful cross-party risk discussion while keeping sensitive evidence within controlled, purpose-limited channels.

IR-079

STAKEHOLDER QUESTION

Which owner-approved evidence threshold should trigger notification, specialist attendance or escalation, and what is the safe response when evidence is uncertain or unavailable?

SENSORS AND TELEMETRY

Proposed evidence: The qualified signal, source-health, corroboration and threshold metadata needed for the named trigger, plus role-based notification-path status; no autonomous control action is included. **Coverage/timing:** The exact asset, zone, hazard and operating mode covered by each owner-approved rule; evaluate at native event timing and retain threshold, suppression and maintenance windows. **Quality gate:** for every contributing stream, verify asset and location mapping, units, calibration or functional check, sampling and aggregation, clock offset, uptime, gaps, drift and source-to-result lineage; exclude any stream that fails its gate. Threshold crossings can be noise, persistent slow change may not cross a limit and failed sensors can create false normal or false alarm states; emergency procedures remain owner-controlled.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: The accountable hazard or control owner, estates duty manager, fire, security or safety owner, competent specialist, incident management and telemetry service owner. Current owner-approved procedures and competent limits lead; qualified telemetry supports trigger evaluation only. **Permissions/provenance:** Retain rule version, asset mapping, unit, hysteresis, corroboration and escalation lineage, restrict sensitive contacts and show any conflict between limits or procedures. **Time/staleness/conflicts:** retain owner-record effective, event observed and query retrieved times; live status is stale after a missed heartbeat or 15 minutes unless the owner sets a stricter limit; preserve zone and time disagreements.

ANALYSIS REQUIRED

Operation type: calculation + comparison + recommendation. **Interval/evidence gate:** Before enabling any advisory trigger; require current accountable approval, exact scope, minimum corroboration, known false-alert basis, named escalation path and tested fallback for source failure. **Method:** Evidence-gated trigger, hysteresis and safe-failure analysis. Evaluate threshold magnitude, persistence, rate of change, corroboration and sensor health, apply hysteresis and classify confirmed trigger, indicated condition or unavailable evidence. **Uncertainty/safe failure:** Report confidence, failed evidence gates and last known good state; default to the owner-defined safe failure and human escalation, never to silent normal.

ANSWER BOUNDARY

The service must not create emergency procedures, dispatch alerts or specialists without separate authority, control equipment, authorise occupation or shutdown, certify danger, or make insurance decisions.

WHY THIS IS USEFUL

Helps owners define transparent escalation rules and prevents uncertain or failed telemetry from being presented as reassurance.

IR-080

STAKEHOLDER QUESTION

What bounded risk statement is supportable now, what is confirmed, indicated, unknown, conflicting or restricted, and which accountable action comes next?

SENSORS AND TELEMETRY

Proposed evidence: Only evidence already qualified for the specific question, with live health, mapping, coverage and freshness status carried into the statement; no additional data is collected merely to make the answer sound complete. **Coverage/timing:** The explicitly stated asset, zone, hazard and decision period; each observation retains its native time and every summary states the coverage window. **Quality gate:** where telemetry is not applicable, validate identifiers, scope, status codes, signatures, timestamp and time-zone alignment, retrieval completeness, missing records, source availability, schema drift and transformation lineage; mark calibration and sampling not applicable. Brevity can hide disagreement, unavailable restricted evidence may limit confidence and a current observation does not establish future performance or policy response.

AUTHORITATIVE SUPPORTING DATA

Owners/precedence: The accountable building and control owners, competent assessors, source custodians, information governance, policy or claims custodian and the intended recipient. Policy terms, competent findings and current accountable-owner records lead within scope; qualified telemetry supports observed conditions and models support scenarios only. **Permissions/provenance:** Preserve claim-to-source lineage, permission and audience, expose conflicts and restrictions, and name the owner of every unresolved material fact and next action. **Time/staleness/conflicts:** retain permission or source effective, evidence observed and query retrieved times; apply freshness to each fact and permission, not the file as a whole; preserve competing values and the accountable owner needed to resolve them.

ANALYSIS REQUIRED

Operation type: lookup + observation + comparison + recommendation. **Interval/evidence gate:** For every delivered answer; require a precise question, authoritative sources, qualified telemetry where used, minimum-evidence gates, uncertainty, answer boundary and a named accountable next owner. **Method:** Bounded claim synthesis and decision-ready risk communication. Assemble confirmed, indicated, unknown, conflicting, restricted and unavailable elements, rank only by stated evidence strength and produce a short statement plus supporting evidence table. **Uncertainty/safe failure:** State coverage, age, uncertainty and scenario limits numerically where possible; safe-fail to 'cannot answer reliably' and the smallest useful evidence request when a material gate fails.

ANSWER BOUNDARY

The service must not recommend or make underwriting, pricing, coverage or claims decisions, infer personal risk or health, certify controls or compliance, expose restricted records, authorise work, or promise outcomes.

WHY THIS IS USEFUL

Provides a usable, challengeable briefing that directs the insurer, assessor and owner to the next accountable evidence or survey action without overstating certainty.

- 1

State the decision-facing question
Name the asset, zone, hazard, audience and decision period. Do not start by collecting everything.
- 2

Confirm authority and permission
Identify source owner, precedence, purpose and access basis before retrieval.
- 3

Qualify coverage and time
Verify mapping, commissioning, health, sampling, effective time, observed time and retrieval time.
- 4

Apply the evidence gate
Use the stated method, quantify missing coverage and uncertainty, and stop safely when a gate fails.
- 5

Give a bounded next action
Separate confirmed, indicated and unknown elements; name the accountable owner, survey or evidence request.

WHAT THE 80 QUESTIONS COVER

- 1

assets, uses, occupancy context and exposure
IR-001 to IR-010
- 2

hazards, vulnerabilities and control ownership
IR-011 to IR-020
- 3

fire, security, water, machinery and environmental risks
IR-021 to IR-030
- 4

incidents, losses, near misses and claims evidence
IR-031 to IR-040
- 5

inspection, maintenance, compliance and impairments
IR-041 to IR-050
- 6

resilience, continuity, dependencies and recovery
IR-051 to IR-060
- 7

climate, weather, energy and long-term change
IR-061 to IR-070
- 8

evidence quality, uncertainty, scenarios and risk communication
IR-071 to IR-080

MINIMUM ANSWER PACKAGE

- Question, asset or zone, audience and decision period
 - Authoritative source owner, scope, version and precedence
 - Effective, observed and retrieved times
 - Sensor mapping, coverage, health, units and quality limits
 - Operation type, evidence gate and method
 - Quantified uncertainty, conflicts and safe failure
 - Answer boundary and named accountable next action
- Missing material elements reduce confidence or stop the answer.**

SOURCE PRECEDENCE WITHIN STATED SCOPE

No source leads outside the fact, time period and scope it is authorised to represent.

1 **Policy terms and formal authoritative records**
Lead for contractual scope, declared facts and accountable decisions.

2 **Competent professional certificates and findings**
Lead for the technical condition and scope they explicitly address.

3 **Current accountable-owner records**
Lead for declared assets, uses, controls, responsibilities and arrangements.

4 **Commissioned and healthy telemetry**
Supports observed physical or system state at a stated place and time.

5 **Analytics, models and scenarios**
Support comparison and bounded reasoning; they do not become authoritative facts or decisions.

MANDATORY EVIDENCE RECORD

Preserve question and scope; source owner and version; effective, observed and retrieved times; asset and location mapping; units and sampling; permission and purpose; transformations; quality and completeness; staleness rule; conflicts; method and assumptions; uncertainty; answer boundary; and named next owner.

RULE 1 - PROPOSED MEANS UNVERIFIED

Never describe a sensor, integration or analytic as deployed until asset and location mapping, commissioning, permissions and current health are independently confirmed.

RULE 2 - PURPOSE BEFORE ACCESS

State the question, audience, decision period and minimum necessary fields before retrieving data. Prior access does not create permission for a new purpose.

RULE 3 - MAP BEFORE INTERPRETING

Every point needs a verified asset, zone, unit, sampling interval and owner. An unmapped signal cannot support a location-specific conclusion.

RULE 4 - PRESERVE THREE TIMES

Retain source effective time, physical or system observed time and query retrieved time. Keep clock and time-zone quality when sequence matters.

RULE 5 - PROVENANCE SURVIVES TRANSFORMATION

Keep the original source reference, extraction parameters, filtering, aggregation, joins, model or rule version and output timestamp. Derived evidence must remain traceable to every input.

RULE 6 - STALENESS IS PURPOSE-SPECIFIC

A certificate may remain current for its stated period while an operational status can become stale within minutes. Apply the accountable owner's freshness rule to the exact fact claimed.

RULE 7 - CONFLICTS REMAIN VISIBLE

Show competing values, source authority, timestamps and affected conclusions. Do not resolve disagreement automatically where an accountable owner or competent professional is required.

RULE 8 - SAFE FAILURE IS AN ANSWER

Use unknown, conflicting, restricted, unavailable or cannot answer reliably when minimum evidence is not met. State the smallest useful next action and responsible owner.

STOP AND ESCALATE WHEN

- A life-safety, water, electrical, security or machinery signal indicates an urgent condition under an existing owner-approved procedure.
- The current authoritative owner cannot be identified, a critical source conflicts, or permission for the intended purpose is absent.
- Sensor mapping, commissioning, calibration or health is unverified, or relevant locations and time periods are materially unobserved.
- A conclusion would require personal-risk inference, restricted disclosure, system control, legal interpretation, certification or a consequential insurance decision.
- The method would extrapolate beyond supported operating states, weather, asset configuration or observed evidence without a competent specialist.

CONSEQUENTIAL DECISIONS REMAIN OUTSIDE THE SERVICE

Policy terms, accountable owners and competent professionals retain authority. The service must not make or recommend underwriting, pricing, coverage or claims decisions; infer personal risk or health; certify controls, condition or compliance; expose restricted records; authorise work, access, occupation or system operation; control equipment; promise response, resilience, loss prevention or recovery; or replace physical surveys, specialist investigation and accountable judgement. Catalogue complete: IR-001 through IR-080.